

PBSAI Control Mapping to NIST SP 800-160 Volume 2

Deterministic Governance and Trustworthy Systems Security Engineering Alignment

Abstract

The Practitioner's Blueprint for Secure AI (PBSAI) Governance Ecosystem is a deterministic governance and operational assurance architecture designed to govern, constrain, validate, authorize, monitor, and audit AI-enabled enterprise systems and autonomous operational workflows.

This document provides a comprehensive mapping between PBSAI governance controls and the systems security engineering techniques and approaches described in NIST Special Publication 800-160 Volume 2.

The mapping demonstrates how PBSAI operationalizes trustworthy secure systems engineering through:

- deterministic runtime governance,
- bounded operational authority,
- explainable execution mediation,
- trusted operational state preservation,
- adaptive governance orchestration,
- resilient operational continuity,
- provenance-aware governance,
- evidence-centric lifecycle management,
- cross-domain telemetry correlation,
- and reconstructable operational accountability.

Unlike traditional governance frameworks focused primarily on organizational policy or static compliance processes, PBSAI introduces runtime operational governance mechanisms specifically designed for:

- AI-enabled systems,
- autonomous orchestration environments,
- multi-agent operational ecosystems,
- adaptive enterprise workflows,
- and continuously governed operational execution.

This document establishes traceable engineering relationships between PBSAI governance controls and NIST SP 800-160 Volume 2 systems security engineering techniques and approaches.

1. Introduction

AI-enabled enterprise systems introduce significant systems-security engineering challenges associated with:

- trustworthy operational behavior,
- bounded autonomy,
- explainable execution,
- runtime controllability,
- resilient orchestration,
- operational survivability,
- trusted state management,
- and adaptive governance enforcement.

Traditional enterprise governance architectures frequently assume:

- human-mediated operational control,
- static authorization boundaries,
- organizationally scoped governance,
- and post-execution auditing.

AI-enabled and autonomous operational systems instead require:

- deterministic runtime governance,
- contextual execution authorization,
- invariant-preserving operational control,
- explainable orchestration constraints,
- bounded execution authority,
- provenance-aware operational governance,
- and continuous governance mediation prior to execution and state transition.

The Practitioner's Blueprint for Secure AI (PBSAI) addresses these requirements through a structured governance architecture spanning:

1. Governance, Risk, and Compliance
2. Change and Configuration Governance
3. Identity and Authorization Governance
4. Threat Intelligence and Situational Awareness
5. Protective Enforcement and Hardening
6. Data Security and Privacy
7. Incident Response and Forensics
8. Resilience and Continuity
9. Security Architecture and Systems Engineering
10. Physical and Environmental Security
11. Supply Chain and Lifecycle Security
12. Security Program and Operational Assurance

This document maps PBSAI governance controls to relevant systems security engineering techniques and approaches described in NIST SP 800-160 Volume 2.

2. Systems Security Engineering Context

NIST SP 800-160 Volume 2 focuses on engineering trustworthy secure systems capable of operating under conditions of:

- complexity,
- uncertainty,
- adversarial pressure,
- operational disruption,
- dynamic environmental change,
- and evolving trust conditions.

PBSAI aligns strongly with systems-security engineering concepts associated with:

- trustworthy autonomous behavior,

- deterministic runtime governance,
- bounded operational authority,
- secure default operational behavior,
- trusted state preservation,
- adaptive governance orchestration,
- resilient execution control,
- explainable operational behavior,
- and reconstructable operational accountability.

PBSAI operationalizes many systems-security engineering concepts directly at runtime through:

- deterministic governance enforcement,
- telemetry-aware orchestration,
- provenance-aware operational control,
- structured execution mediation,
- continuous assurance validation,
- and evidence-centric governance workflows.

3. Scope and Methodology

This mapping aligns PBSAI governance controls to relevant systems security engineering techniques and approaches described in NIST SP 800-160 Volume 2.

Mappings are evaluated using:

- engineering equivalence,
- operational realization,
- runtime governance alignment,
- resilience contribution,
- explainability support,
- trustworthy systems engineering semantics,
- and operational assurance contribution.

Mappings focus particularly on PBSAI support for:

- trustworthy AI-enabled systems,
 - resilient runtime governance,
 - bounded operational authority,
 - deterministic execution governance,
 - trusted operational state preservation,
 - explainable orchestration,
 - provenance-aware governance,
 - evidence-centric operational accountability,
 - and continuously adaptive operational governance.
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4. Relationship Semantics

Relationship	Definition
equivalent	PBSAI materially realizes the systems-security engineering technique and approach
strongly-supports	PBSAI substantially enables the engineering objective
supports	PBSAI partially contributes to the engineering objective
operationalizes	PBSAI converts engineering guidance into runtime governance enforcement
constrains	PBSAI enforces runtime engineering boundaries
extends	PBSAI introduces governance capabilities beyond the engineering guidance

5. PBSAI Domain A Controls → NIST SP 800-160 Volume 2

A-GOV-001 — Governance Policy Management

Technique	Approach	Relationship	Engineering Rationale
Governance Authority Engineering	Trusted Operational Governance	equivalent	Establishes authoritative governance policy structures for AI-enabled operational systems
Deterministic Policy Enforcement	Runtime Governance Mediation	strongly-supports	Enables runtime enforcement of governance policy constraints
Accountability Preservation	Evidence-Centric Operations	supports	Preserves attributable governance-policy lineage and operational accountability

A-GOV-002 — Governance Roles and Responsibilities

Technique	Approach	Relationship	Engineering Rationale
Governance Accountability Engineering	Accountability Preservation	equivalent	Defines attributable operational governance responsibilities
Supervisory Governance Modeling	Human-Governed Operations	strongly-supports	Preserves authoritative governance oversight structures
Role-Constrained Authorization	Trusted Operational Control	supports	Restricts operational authority according to governed responsibility boundaries

A-RISK-001 — AI Risk Governance

Technique	Approach	Relationship	Engineering Rationale
Dynamic Risk Governance	Adaptive Governance	equivalent	Integrates runtime operational risk into governance evaluation and operational control
Risk-Aware Operational Mediation	Safe Operational Behavior	strongly-supports	Applies governance restrictions according to operational risk posture

Technique	Approach	Relationship	Engineering Rationale
Continuous Trust Assessment	Resilient Operational Awareness	supports	Maintains visibility into evolving operational trust conditions

A-RISK-002 — Enterprise Risk Assessment

Technique	Approach	Relationship	Engineering Rationale
Enterprise Risk Evaluation	Adaptive Governance	equivalent	Evaluates operational exposure across enterprise AI-enabled environments
Cross-Domain Risk Correlation	Resilient Operational Awareness	strongly-supports	Correlates operational risk conditions across governance domains
Exposure Prioritization	Trusted Operational Control	supports	Enables governance prioritization based on operational criticality

A-RISK-003 — Risk Prioritization and Treatment

Technique	Approach	Relationship	Engineering Rationale
Risk-Adaptive Governance	Resilient Operational Control	equivalent	Applies operational mitigation according to contextual risk severity
Operational Treatment Coordination	Adaptive Governance	strongly-supports	Coordinates governance mitigation across operational domains
Constraint-Aware Risk Reduction	Safe Operational Behavior	supports	Preserves trusted operational conditions during mitigation workflows

6. PBSAI Domain B Controls → NIST SP 800-160 Volume 2

B-ASSET-001 — Asset Inventory Governance

Technique	Approach	Relationship	Engineering Rationale
Trusted Asset Visibility	Trusted Operational Awareness	equivalent	Maintains authoritative visibility into governed enterprise operational assets
Operational Dependency Awareness	Resilient Operational Control	strongly-supports	Enables dependency-aware governance and operational risk assessment
Lifecycle Traceability Engineering	Evidence-Centric Operations	supports	Preserves attributable operational asset lineage and governance continuity

B-CONFIG-001 — Configuration Baseline Governance

Technique	Approach	Relationship	Engineering Rationale
Deterministic Configuration Governance	Trusted Operational State	equivalent	Preserves governed baseline operational configurations
Configuration Integrity Preservation	Runtime Governance Mediation	strongly-supports	Prevents unauthorized or unsafe configuration divergence
Controlled Operational State Management	Safe Operational Behavior	supports	Maintains trusted operational execution conditions

B-CHANGE-001 — Change Governance and Tracking

Technique	Approach	Relationship	Engineering Rationale
Controlled Operational Change Management	Trusted Operational Control	equivalent	Governs operational modifications through deterministic change mediation
Traceable Change Accountability	Accountability Preservation	strongly-supports	Preserves attributable operational change history

Technique	Approach	Relationship	Engineering Rationale
Explainable Operational Transition	Explainable Operational Behavior	supports	Enables reconstructable analysis of governed operational modifications

B-CHANGE-002 — Change Authorization Validation

Technique	Approach	Relationship	Engineering Rationale
Authorization Integrity Enforcement	Governance Authority Preservation	equivalent	Validates authoritative approval prior to operational modification
Runtime Approval Enforcement	Human-Governed Operations	strongly-supports	Prevents unauthorized operational state transition
Bounded Operational Governance	Deterministic Governance	supports	Restricts operational changes to approved governance boundaries

B-CHANGE-003 — Configuration Drift Detection

Technique	Approach	Relationship	Engineering Rationale
Runtime Integrity Monitoring	Trusted Operational Assurance	equivalent	Detects unauthorized divergence from governed operational baseline
Drift-Aware Operational Governance	Adaptive Governance	strongly-supports	Enables adaptive governance response to operational deviation
Continuous State Validation	Trusted Operational State	supports	Maintains visibility into trusted operational conditions

B-CHANGE-004 — Rollback Governance

Technique	Approach	Relationship	Engineering Rationale
Recovery-Aware Governance	Survivability Engineering	equivalent	Enables safe restoration following failed or unsafe operational modifications
Trusted State Restoration	Resilient Operational Control	strongly-supports	Preserves trusted operational continuity during rollback workflows
Safe Operational Recovery	Safe Operational Behavior	supports	Prevents unsafe persistence of invalid operational configurations

B-DEPLOY-001 — Deployment Governance

Technique	Approach	Relationship	Engineering Rationale
Governed Operational Deployment	Trusted Operational Control	equivalent	Applies deterministic governance prior to operational deployment
Controlled State Transition	Runtime Governance Mediation	strongly-supports	Restricts deployment execution to governed authorization boundaries
Operational Integrity Preservation	Trusted Operational State	supports	Preserves trusted execution conditions during deployment activities

B-DEPLOY-002 — Deployment Validation and Verification

Technique	Approach	Relationship	Engineering Rationale
Trusted Operational Verification	Trusted Operational Assurance	equivalent	Validates deployment integrity and operational correctness prior to activation
Runtime Integrity Assurance	Safe Operational Behavior	strongly-supports	Prevents unsafe operational deployment conditions

Technique	Approach	Relationship	Engineering Rationale
Evidence-Aware Deployment Governance	Evidence-Centric Operations	supports	Preserves attributable deployment-validation evidence continuity

B-EVID-001 — Configuration and Change Audit Logging

Technique	Approach	Relationship	Engineering Rationale
Operational Event Logging	Evidence-Centric Operations	equivalent	Maintains authoritative records of configuration and change activities
Lifecycle Accountability Preservation	Accountability Preservation	strongly-supports	Preserves attributable operational change accountability
Explainable Operational Reconstruction	Explainable Operational Behavior	supports	Enables reconstructable analysis of governed operational transitions

B-STATE-001 — Operational State Continuity

Technique	Approach	Relationship	Engineering Rationale
Trusted State Preservation	Trusted Operational State	equivalent	Maintains continuity of governed operational execution state
State Integrity Continuity	Resilient Operational Control	strongly-supports	Prevents undefined or inconsistent operational conditions
Operational Recovery Coordination	Survivability Engineering	supports	Enables recoverable operational continuity during disruption conditions

B-AUTO-001 — Automated Configuration Governance

Technique	Approach	Relationship	Engineering Rationale
Deterministic Automation Governance	Runtime Governance Mediation	equivalent	Applies deterministic governance to automated configuration workflows
Constraint-Aware Automation	Safe Operational Behavior	strongly-supports	Prevents unsafe autonomous operational modification
Governed Orchestration Control	Adaptive Governance	supports	Preserves bounded automation authority during runtime operations

7. PBSAI Domain C Controls → NIST SP 800-160 Volume 2

C-ID-001 — Identity Governance

Technique	Approach	Relationship	Engineering Rationale
Trusted Identity Establishment	Trusted Operational Interaction	equivalent	Maintains authoritative operational identity governance
Identity Accountability Preservation	Accountability Preservation	strongly-supports	Preserves attributable operational identity continuity
Runtime Identity Assurance	Trusted Operational State	supports	Maintains trusted identity conditions during execution lifecycle

C-AUTH-001 — Authorization Policy Enforcement

Technique	Approach	Relationship	Engineering Rationale
Deterministic Authorization Enforcement	Trusted Operational Control	equivalent	Enforces policy-bound authorization prior to operational execution
Runtime Governance Mediation	Constraint-Aware Governance	strongly-supports	Preserves governed operational authorization boundaries

Technique	Approach	Relationship	Engineering Rationale
Least-Authority Execution	Bounded Operational Authority	supports	Restricts execution authority to approved operational scope

C-ROLE-001 — Role and Privilege Governance

Technique	Approach	Relationship	Engineering Rationale
Least Privilege Enforcement	Bounded Operational Authority	equivalent	Restricts operational privileges to governed authorization boundaries
Privilege Integrity Preservation	Trusted Operational Governance	strongly-supports	Prevents unauthorized privilege escalation
Role Accountability Traceability	Evidence-Centric Operations	supports	Preserves attributable operational role lineage

C-SESS-001 — Session and Context Continuity Governance

Technique	Approach	Relationship	Engineering Rationale
Context Continuity Preservation	Trusted Operational State	equivalent	Maintains authoritative session continuity across operational workflows
Runtime Context Propagation	Adaptive Operational Awareness	strongly-supports	Preserves governance-aware contextual execution conditions
Traceable Operational Sessions	Accountability Preservation	supports	Enables attributable reconstruction of session-governed operational behavior

C-MFA-001 — Multi-Factor Authentication Enforcement

Technique	Approach	Relationship	Engineering Rationale
Strong Identity Assurance	Trusted Operational Interaction	equivalent	Applies elevated identity validation prior to operational authorization
Compromise-Resistant Authentication	Trusted Operational Control	strongly-supports	Reduces operational exposure from identity compromise
Adaptive Identity Protection	Resilient Operational Awareness	supports	Enables risk-aware authentication governance workflows

C-ACCESS-001 — Adaptive Access Governance

Technique	Approach	Relationship	Engineering Rationale
Context-Aware Authorization	Adaptive Governance	equivalent	Dynamically adjusts operational authorization according to contextual risk
Runtime Access Mediation	Runtime Governance Mediation	strongly-supports	Preserves deterministic access governance during execution lifecycle
Risk-Adaptive Operational Control	Safe Operational Behavior	supports	Prevents unsafe operational access conditions

C-TRACE-001 — Identity-to-Action Traceability

Technique	Approach	Relationship	Engineering Rationale
Traceability Engineering	Accountability Preservation	equivalent	Maintains attributable operational execution traceability
Evidence Binding	Evidence-Centric Operations	strongly-supports	Preserves attributable governance continuity

Technique	Approach	Relationship	Engineering Rationale
Explainable Operational Reconstruction	Explainable Operational Behavior	supports	Enables reconstructable operational accountability analysis

8. PBSAI Domain D Controls → NIST SP 800-160 Volume 2

D-THREAT-001 — Threat Intelligence Integration

Technique	Approach	Relationship	Engineering Rationale
Threat-Aware Operational Governance	Adaptive Operational Awareness	equivalent	Integrates evolving threat intelligence into operational governance evaluation
Dynamic Adversarial Awareness	Resilient Operational Control	strongly- supports	Maintains visibility into changing operational threat conditions
Contextual Threat Correlation	Explainable Operational Behavior	supports	Associates operational risk conditions with threat- intelligence context

D-THREAT-002 — Threat Indicator Correlation

Technique	Approach	Relationship	Engineering Rationale
Cross-Domain Threat Correlation	Trusted Operational Awareness	equivalent	Correlates indicators across operational governance domains
Adversarial Pattern Detection	Survivability Engineering	strongly- supports	Enables early detection of coordinated operational threats
Evidence-Aware Threat Reconstruction	Evidence-Centric Operations	supports	Preserves attributable threat-analysis continuity

D-THREAT-003 — Threat-Adaptive Monitoring

Technique	Approach	Relationship	Engineering Rationale
Adaptive Monitoring Governance	Adaptive Governance	equivalent	Dynamically adjusts operational monitoring according to evolving threats
Runtime Threat Mediation	Runtime Governance Mediation	strongly-supports	Enables governance adaptation during changing adversarial conditions
Continuous Trust Evaluation	Trusted Operational Assurance	supports	Maintains operational trust visibility under dynamic threat conditions

D-THREAT-004 — Threat Severity Prioritization

Technique	Approach	Relationship	Engineering Rationale
Risk-Aware Threat Governance	Trusted Operational Control	equivalent	Prioritizes operational response according to threat severity and exposure
Operational Exposure Evaluation	Resilient Operational Awareness	strongly-supports	Correlates threat conditions with enterprise operational criticality
Constraint-Aware Mitigation	Safe Operational Behavior	supports	Enables bounded operational response prioritization

D-THREAT-005 — Threat Intelligence Feedback Integration

Technique	Approach	Relationship	Engineering Rationale
Adaptive Threat Governance	Adaptive Governance	equivalent	Integrates evolving threat intelligence into operational governance refinement
Continuous Adversarial Learning	Survivability Engineering	strongly-supports	Enables adaptive operational resilience against evolving threats

Technique	Approach	Relationship	Engineering Rationale
Governance Optimization Feedback	Runtime Governance Mediation	supports	Improves operational governance effectiveness through threat-informed adaptation

D-SA-001 — Situational Awareness Correlation

Technique	Approach	Relationship	Engineering Rationale
Cross-Domain Situational Awareness	Trusted Operational Awareness	equivalent	Correlates telemetry and operational state across enterprise domains
Operational Context Reconstruction	Explainable Operational Behavior	strongly-supports	Enables explainable interpretation of operational conditions
Runtime Operational Visibility	Trusted Operational State	supports	Maintains awareness of evolving operational trust conditions

D-SA-002 — Behavioral and Operational Correlation

Technique	Approach	Relationship	Engineering Rationale
Behavioral Analytics Engineering	Adaptive Operational Awareness	equivalent	Detects anomalous operational behavior across enterprise workflows
Pattern-Based Threat Detection	Survivability Engineering	strongly-supports	Enables identification of abnormal operational activity
Cross-Domain Event Fusion	Evidence-Centric Operations	supports	Preserves attributable operational correlation continuity

D-MON-001 — Continuous Monitoring

Technique	Approach	Relationship	Engineering Rationale
Continuous Operational Observation	Trusted Operational Assurance	equivalent	Continuously monitors AI-enabled enterprise operational conditions
Runtime Governance Visibility	Runtime Governance Mediation	strongly-supports	Maintains governance awareness during operational execution
Operational Drift Detection	Adaptive Governance	supports	Detects divergence from trusted operational conditions

D-MON-002 — Telemetry Integrity Validation

Technique	Approach	Relationship	Engineering Rationale
Trusted Telemetry Assurance	Trusted Operational State	equivalent	Validates integrity and trustworthiness of operational telemetry
Evidence Integrity Preservation	Evidence-Centric Operations	strongly-supports	Prevents corruption or falsification of operational evidence
Operational Monitoring Reliability	Resilient Operational Control	supports	Maintains dependable situational awareness conditions

D-EVID-001 — Security Event Logging

Technique	Approach	Relationship	Engineering Rationale
Operational Evidence Collection	Evidence-Centric Operations	equivalent	Produces attributable security-event telemetry and governance evidence
Explainable Operational Reconstruction	Explainable Operational Behavior	strongly-supports	Enables reconstructable analysis of operational activity

Technique	Approach	Relationship	Engineering Rationale
Accountability Preservation	Accountability Preservation	supports	Maintains attributable operational governance history

D-EVID-002 — Telemetry Retention and Preservation

Technique	Approach	Relationship	Engineering Rationale
Operational Evidence Retention	Trusted Operational Assurance	equivalent	Preserves operational telemetry continuity for governance reconstruction
Forensic Continuity Engineering	Survivability Engineering	strongly-supports	Maintains reconstructable operational evidence during disruption conditions
Longitudinal Governance Visibility	Adaptive Operational Awareness	supports	Enables historical analysis of evolving operational trust conditions

D-ANL-001 — Security Analytics and Operational Assessment

Technique	Approach	Relationship	Engineering Rationale
Operational Analytics Engineering	Trusted Operational Awareness	equivalent	Produces governance-aware operational security analytics
Continuous Exposure Evaluation	Adaptive Governance	strongly-supports	Enables dynamic operational risk and trust assessment
Explainable Operational Assessment	Explainable Operational Behavior	supports	Preserves interpretability of governance analytics and operational evaluation

9. PBSAI Domain E Controls → NIST SP 800-160 Volume 2

E-PRO-001 — Preventive Control Definition

Technique	Approach	Relationship	Engineering Rationale
Preventive Governance Engineering	Safe Operational Behavior	equivalent	Establishes deterministic preventive operational controls
Runtime Constraint Enforcement	Runtime Governance Mediation	strongly-supports	Applies governance restrictions prior to operational execution
Bounded Operational Authority	Trusted Operational Control	supports	Prevents unsafe operational execution conditions

E-PRO-002 — Segmentation and Isolation Enforcement

Technique	Approach	Relationship	Engineering Rationale
Segmentation-Aware Governance	Trusted Operational State	equivalent	Enforces operational isolation and trust-boundary protection
Exposure Reduction Engineering	Survivability Engineering	strongly-supports	Limits propagation of operational compromise conditions
Constraint-Aware Operational Separation	Safe Operational Behavior	supports	Preserves bounded operational interaction pathways

E-PRO-003 — Endpoint and Workload Protection

Technique	Approach	Relationship	Engineering Rationale
Runtime Operational Protection	Trusted Operational Assurance	equivalent	Protects enterprise systems and workloads during execution lifecycle
Operational Integrity Preservation	Trusted Operational State	strongly-supports	Maintains trustworthy runtime execution conditions

Technique	Approach	Relationship	Engineering Rationale
Adaptive Protection Coordination	Resilient Operational Control	supports	Enables evolving protection governance during changing operational conditions

E-PRO-004 — Protected Data Access Enforcement

Technique	Approach	Relationship	Engineering Rationale
Data Access Governance	Trusted Operational Control	equivalent	Restricts governed access to protected enterprise data resources
Runtime Authorization Enforcement	Deterministic Governance	strongly-supports	Prevents unauthorized operational data access
Privacy-Aware Operational Governance	Safe Operational Behavior	supports	Preserves controlled handling of sensitive operational information

E-ENF-001 — Policy Enforcement Governance

Technique	Approach	Relationship	Engineering Rationale
Deterministic Enforcement Governance	Runtime Governance Mediation	equivalent	Applies runtime policy mediation to operational workflows
Trusted Operational Constraint Enforcement	Safe Operational Behavior	strongly-supports	Prevents execution outside governed operational boundaries
Explainable Enforcement Continuity	Explainable Operational Behavior	supports	Preserves attributable operational enforcement traceability

E-ENF-002 — Response Authorization Validation

Technique	Approach	Relationship	Engineering Rationale
Authorized Response Governance	Human-Governed Operations	equivalent	Validates authoritative approval prior to operational response execution
Governance-Bound Remediation	Deterministic Governance	strongly-supports	Restricts remediation workflows to governed authorization scope
Trusted Operational Intervention	Trusted Operational Control	supports	Preserves safe operational remediation conditions

E-ENF-003 — Response Scope Constraint

Technique	Approach	Relationship	Engineering Rationale
Bounded Remediation Governance	Safe Operational Behavior	equivalent	Restricts remediation workflows to approved operational scope
Runtime Constraint Preservation	Deterministic Governance	strongly-supports	Prevents excessive or unsafe operational enforcement
Controlled Operational Recovery	Survivability Engineering	supports	Maintains trusted operational continuity during mitigation workflows

E-AUTO-001 — Automated Enforcement Governance

Technique	Approach	Relationship	Engineering Rationale
Governed Autonomous Enforcement	Adaptive Governance	equivalent	Applies deterministic governance to automated operational enforcement
Constraint-Aware Automation	Runtime Governance Mediation	strongly-supports	Preserves bounded automation authority during runtime operations

Technique	Approach	Relationship	Engineering Rationale
Trusted Autonomous Response	Safe Operational Behavior	supports	Prevents unsafe autonomous operational remediation

E-EVID-001 — Enforcement Event Logging

Technique	Approach	Relationship	Engineering Rationale
Enforcement Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable operational enforcement telemetry
Operational Accountability Continuity	Accountability Preservation	strongly-supports	Preserves reconstructable governance enforcement lineage
Explainable Enforcement Reconstruction	Explainable Operational Behavior	supports	Enables reconstructable analysis of operational enforcement activities

E-SEG-001 — Dynamic Segmentation Governance

Technique	Approach	Relationship	Engineering Rationale
Adaptive Segmentation Control	Resilient Operational Control	equivalent	Dynamically adjusts operational segmentation according to runtime conditions
Runtime Trust-Boundary Preservation	Trusted Operational State	strongly-supports	Maintains trusted operational separation during evolving threat conditions
Survivable Operational Isolation	Survivability Engineering	supports	Limits operational compromise propagation during adaptive response workflows

10. PBSAI Domain F Controls → NIST SP 800-160 Volume 2

F-CLS-001 — Data Discovery

Technique	Approach	Relationship	Engineering Rationale
Trusted Data Asset Visibility	Trusted Operational Awareness	equivalent	Establishes authoritative visibility into governed enterprise data assets
Data Context Engineering	Adaptive Operational Awareness	strongly-supports	Enables governance decisions based on data location, sensitivity, and operational context
Evidence-Aware Data Mapping	Evidence-Centric Operations	supports	Preserves attributable visibility into governed data discovery activities

F-CLS-002 — Data Classification

Technique	Approach	Relationship	Engineering Rationale
Information Classification Engineering	Trusted Operational State	equivalent	Assigns governed classification attributes to data assets
Contextual Data Governance	Runtime Governance Mediation	strongly-supports	Enables policy evaluation based on data sensitivity and handling requirements
Attribute-Based Control Support	Safe Operational Behavior	supports	Reduces unsafe use of sensitive or regulated data

F-CLS-003 — Classification Validation

Technique	Approach	Relationship	Engineering Rationale
Trusted Information Validation	Trusted Operational Assurance	equivalent	Validates correctness of data classification decisions
Classification Integrity Assurance	Trusted Operational State	strongly-supports	Prevents governance decisions based on incorrect data sensitivity attributes

Technique	Approach	Relationship	Engineering Rationale
Continuous Data Trust Assessment	Adaptive Governance	supports	Enables adaptive correction of classification drift or mislabeling

F-CLS-004 — Classification Metadata Propagation

Technique	Approach	Relationship	Engineering Rationale
Trusted Attribute Propagation	Runtime Governance Mediation	equivalent	Preserves classification metadata across operational workflows
Context Continuity Preservation	Trusted Operational State	strongly-supports	Maintains data-governance context during cross-domain processing
Traceable Data Handling	Evidence-Centric Operations	supports	Enables reconstruction of classification-aware data operations

F-CLS-005 — Classification-Based Policy Binding

Technique	Approach	Relationship	Engineering Rationale
Policy-Bound Data Governance	Trusted Operational Control	equivalent	Binds operational data handling to classification-driven policy constraints
Context-Aware Authorization	Runtime Governance Mediation	strongly-supports	Applies differentiated governance based on data sensitivity
Safe Information Handling	Safe Operational Behavior	supports	Reduces unsafe operational use of protected data

F-DLP-001 — Data Movement Monitoring

Technique	Approach	Relationship	Engineering Rationale
Information Flow Monitoring	Trusted Operational Awareness	equivalent	Monitors governed data movement across systems and boundaries
Runtime Data Flow Visibility	Adaptive Operational Awareness	strongly-supports	Enables detection of anomalous or unauthorized data transfers
Evidence-Aware Flow Reconstruction	Evidence-Centric Operations	supports	Preserves traceable records of data movement activity

F-DLP-002 — Unauthorized Data Movement Detection

Technique	Approach	Relationship	Engineering Rationale
Information Flow Constraint Detection	Safe Operational Behavior	equivalent	Detects data movement inconsistent with governed flow constraints
Runtime Exfiltration Awareness	Survivability Engineering	strongly-supports	Supports early detection of compromise or unauthorized disclosure pathways
Adaptive Data Protection	Resilient Operational Control	supports	Enables policy-based response to changing data-movement risk

F-DLP-003 — Data Movement Blocking and Containment

Technique	Approach	Relationship	Engineering Rationale
Information Flow Enforcement	Trusted Operational Control	equivalent	Blocks or constrains prohibited data flows
Runtime Boundary Protection	Safe Operational Behavior	strongly-supports	Prevents unsafe or unauthorized movement of sensitive information

Technique	Approach	Relationship	Engineering Rationale
Containment-Aware Data Governance	Survivability Engineering	supports	Limits operational impact of suspected data compromise

F-DLP-004 — High-Impact DLP Escalation

Technique	Approach	Relationship	Engineering Rationale
Supervisory Data Protection Control	Human-Governed Operations	equivalent	Escalates high-impact data protection decisions for human review
Data Risk Escalation Engineering	Governance Accountability	strongly-supports	Preserves accountable oversight for sensitive data-control actions
Safe Operational Intervention	Safe Operational Behavior	supports	Prevents excessive or unsafe automated data enforcement

F-DLP-005 — DLP Evidence Generation

Technique	Approach	Relationship	Engineering Rationale
Data Protection Evidence Collection	Evidence-Centric Operations	equivalent	Produces attributable evidence for data movement and DLP decisions
Data Flow Traceability	Accountability Preservation	strongly-supports	Maintains reconstructable data-protection accountability
Explainable Data Governance	Explainable Operational Behavior	supports	Enables review and reconstruction of DLP enforcement decisions

F-CRY-001 — Encryption at Rest and in Transit

Technique	Approach	Relationship	Engineering Rationale
Cryptographic Information Protection	Secure Information Protection	equivalent	Protects sensitive data during storage and transmission
Confidentiality Preservation	Trusted Operational Communications	strongly-supports	Maintains trusted confidentiality boundaries across operational workflows
Trusted Data State Preservation	Trusted Operational State	supports	Preserves protected operational data conditions

F-CRY-002 — Cryptographic Configuration Compliance

Technique	Approach	Relationship	Engineering Rationale
Cryptographic Configuration Governance	Trusted Operational Control	equivalent	Ensures cryptographic settings conform to governed baselines
Configuration Integrity Assurance	Trusted Operational State	strongly-supports	Prevents weak or unauthorized cryptographic configurations
Runtime Cryptographic Governance	Runtime Governance Mediation	supports	Enables governance evaluation of cryptographic posture

F-CRY-003 — Weak Cryptography Detection

Technique	Approach	Relationship	Engineering Rationale
Cryptographic Weakness Detection	Trusted Operational Assurance	equivalent	Identifies deprecated, weak, or nonconforming cryptographic mechanisms
Adaptive Crypto Risk Evaluation	Adaptive Governance	strongly-supports	Enables governance response to emerging cryptographic weakness

Technique	Approach	Relationship	Engineering Rationale
Operational Exposure Reduction	Survivability Engineering	supports	Reduces compromise risk arising from weak protection mechanisms

F-CRY-004 — Cryptographic Remediation

Technique	Approach	Relationship	Engineering Rationale
Cryptographic Remediation Governance	Resilient Operational Control	equivalent	Coordinates correction of weak or noncompliant cryptographic protections
Controlled Security State Transition	Runtime Governance Mediation	strongly-supports	Ensures crypto remediation occurs through governed change paths
Trusted Protection Restoration	Trusted Operational State	supports	Restores governed protection posture after cryptographic weakness detection

F-CRY-005 — Crypto Evidence and Attestation

Technique	Approach	Relationship	Engineering Rationale
Cryptographic Evidence Preservation	Evidence-Centric Operations	equivalent	Maintains evidence of cryptographic posture, validation, and remediation
Attested Protection State	Trusted Operational Assurance	strongly-supports	Supports verifiable claims regarding cryptographic protection state
Explainable Security Posture	Explainable Operational Behavior	supports	Enables reconstructable review of cryptographic governance decisions

10. PBSAI Domain F Controls → NIST SP 800-160 Volume 2 (continued)

F-INT-001 — Data Integrity Validation

Technique	Approach	Relationship	Engineering Rationale
Trusted Information Integrity Assurance	Trusted Operational State	equivalent	Validates integrity and trustworthiness of governed enterprise data
Runtime Integrity Verification	Trusted Operational Assurance	strongly-supports	Detects corruption or unauthorized modification of operational data
Integrity-Aware Operational Governance	Safe Operational Behavior	supports	Prevents execution based on untrusted or compromised information

F-INT-002 — Integrity Violation Detection

Technique	Approach	Relationship	Engineering Rationale
Integrity Anomaly Detection	Adaptive Operational Awareness	equivalent	Detects operational conditions inconsistent with trusted data integrity
Corruption-Aware Governance	Survivability Engineering	strongly-supports	Enables rapid mitigation of integrity compromise conditions
Runtime Trust Evaluation	Trusted Operational Assurance	supports	Maintains visibility into evolving operational trustworthiness conditions

F-INT-003 — Integrity Enforcement and Restoration

Technique	Approach	Relationship	Engineering Rationale
Integrity Restoration Engineering	Resilient Operational Control	equivalent	Coordinates recovery from compromised or invalid data states
Trusted State Recovery	Trusted Operational State	strongly-supports	Restores trustworthy operational information conditions

Technique	Approach	Relationship	Engineering Rationale
Constraint-Aware Recovery Governance	Safe Operational Behavior	supports	Prevents unsafe operational continuation following integrity compromise

F-INT-004 — Integrity Chain Preservation

Technique	Approach	Relationship	Engineering Rationale
Integrity Lineage Preservation	Evidence-Centric Operations	equivalent	Maintains reconstructable integrity lineage across operational workflows
Trust Continuity Engineering	Trusted Operational Assurance	strongly-supports	Preserves attributable integrity state transitions
Explainable Integrity Reconstruction	Explainable Operational Behavior	supports	Enables forensic reconstruction of integrity degradation conditions

F-INT-005 — Integrity Evidence Generation

Technique	Approach	Relationship	Engineering Rationale
Integrity Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence of integrity validation and enforcement
Accountability-Aware Integrity Governance	Accountability Preservation	strongly-supports	Maintains reconstructable integrity-governance continuity
Explainable Integrity Analysis	Explainable Operational Behavior	supports	Enables explainable review of integrity-assurance decisions

F-PRV-001 — Purpose Limitation Enforcement

Technique	Approach	Relationship	Engineering Rationale
Purpose-Bound Information Governance	Trusted Operational Control	equivalent	Restricts operational data use according to governed processing purpose
Context-Aware Data Authorization	Runtime Governance Mediation	strongly-supports	Applies operational constraints based on lawful processing context
Privacy-Constrained Execution	Safe Operational Behavior	supports	Prevents unsafe or unauthorized operational data use

F-PRV-002 — Lawful Basis and Regulatory Validation

Technique	Approach	Relationship	Engineering Rationale
Regulatory Governance Validation	Governance Accountability	equivalent	Validates lawful authority for operational data processing
Compliance-Aware Operational Governance	Adaptive Governance	strongly-supports	Enables runtime governance adaptation according to regulatory obligations
Trusted Privacy Governance	Trusted Operational Control	supports	Preserves lawful and governed operational information handling

F-PRV-003 — Consent and Authorization Governance

Technique	Approach	Relationship	Engineering Rationale
Consent-Aware Governance	Human-Governed Operations	equivalent	Preserves governed authorization for operational data usage
Runtime Consent Validation	Runtime Governance Mediation	strongly-supports	Prevents processing inconsistent with authorized consent conditions

Technique	Approach	Relationship	Engineering Rationale
Explainable Consent Traceability	Explainable Operational Behavior	supports	Enables reconstructable consent-governance analysis

F-PRV-004 — Sensitive Data Protection

Technique	Approach	Relationship	Engineering Rationale
Sensitive Information Governance	Trusted Operational State	equivalent	Applies elevated protection controls to sensitive operational information
High-Impact Data Risk Reduction	Survivability Engineering	strongly- supports	Reduces exposure arising from compromise of regulated or sensitive data
Safe Information Handling	Safe Operational Behavior	supports	Prevents unsafe operational interaction with sensitive information

F-PRV-005 — Privacy Evidence and Auditability

Technique	Approach	Relationship	Engineering Rationale
Privacy Accountability Preservation	Evidence-Centric Operations	equivalent	Maintains attributable records for privacy-governance decisions
Explainable Privacy Governance	Explainable Operational Behavior	strongly- supports	Enables reconstructable analysis of governed privacy operations
Traceable Data Governance	Accountability Preservation	supports	Preserves operational lineage for sensitive data handling workflows

F-DSP-001 — Retention Enforcement

Technique	Approach	Relationship	Engineering Rationale
Lifecycle-Aware Information Governance	Trusted Operational Control	equivalent	Enforces governed retention and deletion requirements
Runtime Data Lifecycle Enforcement	Runtime Governance Mediation	strongly-supports	Prevents unauthorized retention or persistence of governed data
Controlled Information Disposal	Safe Operational Behavior	supports	Reduces operational exposure from unnecessary retained information

F-DSP-002 — Secure Data Disposal

Technique	Approach	Relationship	Engineering Rationale
Secure Information Destruction	Trusted Operational State	equivalent	Ensures protected destruction of governed enterprise data
Residual Exposure Reduction	Survivability Engineering	strongly-supports	Prevents recovery or persistence of disposed information
Trusted Lifecycle Completion	Trusted Operational Assurance	supports	Maintains trustworthy completion of operational data lifecycle workflows

F-DSP-003 — Retention Exception Governance

Technique	Approach	Relationship	Engineering Rationale
Exception-Aware Governance	Adaptive Governance	equivalent	Applies controlled governance to retention exceptions and overrides
Supervisory Retention Control	Human-Governed Operations	strongly-supports	Preserves accountable oversight for exceptional retention decisions

Technique	Approach	Relationship	Engineering Rationale
Explainable Lifecycle Governance	Explainable Operational Behavior	supports	Enables reconstructable review of retention exception workflows

F-DSP-004 — Data Lifecycle Traceability

Technique	Approach	Relationship	Engineering Rationale
Information Lifecycle Traceability	Evidence-Centric Operations	equivalent	Preserves reconstructable lineage across the operational data lifecycle
Lifecycle Accountability Preservation	Accountability Preservation	strongly-supports	Maintains attributable operational history for governed data
Trusted Lifecycle Awareness	Trusted Operational Awareness	supports	Enables visibility into evolving data lifecycle state

F-EVID-001 — Data Action Evidence Records

Technique	Approach	Relationship	Engineering Rationale
Data Governance Evidence Collection	Evidence-Centric Operations	equivalent	Produces attributable evidence for governed data operations
Operational Data Accountability	Accountability Preservation	strongly-supports	Preserves reconstructable lineage of operational data actions
Explainable Information Governance	Explainable Operational Behavior	supports	Enables explainable review of governed data handling decisions

F-EVID-002 — Data Lineage and Handling Traceability

Technique	Approach	Relationship	Engineering Rationale
Information Lineage Preservation	Trusted Operational Assurance	equivalent	Maintains reconstructable lineage for governed operational data
Traceable Data Flow Governance	Evidence-Centric Operations	strongly-supports	Enables attributable reconstruction of operational information handling
Context-Aware Operational Visibility	Adaptive Operational Awareness	supports	Preserves visibility into evolving operational data conditions

F-EVID-003 — Cross-Domain Data Correlation

Technique	Approach	Relationship	Engineering Rationale
Cross-Domain Information Correlation	Trusted Operational Awareness	equivalent	Correlates governed data events across operational governance domains
Multi-Domain Evidence Fusion	Evidence-Centric Operations	strongly-supports	Enables enterprise-scale operational reconstruction and analysis
Explainable Operational Correlation	Explainable Operational Behavior	supports	Preserves explainable linkage between related operational events

11. PBSAI Domain G Controls → NIST SP 800-160 Volume 2

G-IR-001 — Incident Identification

Technique	Approach	Relationship	Engineering Rationale
Operational Incident Detection	Trusted Operational Awareness	equivalent	Detects operational incidents affecting governed enterprise systems

Technique	Approach	Relationship	Engineering Rationale
Runtime Threat Recognition	Adaptive Operational Awareness	strongly-supports	Enables awareness of evolving operational compromise conditions
Early Operational Disruption Detection	Survivability Engineering	supports	Improves ability to respond before operational degradation propagates

G-IR-002 — Incident Classification and Severity Assessment

Technique	Approach	Relationship	Engineering Rationale
Operational Risk Prioritization	Trusted Operational Control	equivalent	Classifies operational incidents according to impact and severity
Adaptive Incident Governance	Adaptive Governance	strongly-supports	Enables governance response proportional to operational exposure
Explainable Incident Evaluation	Explainable Operational Behavior	supports	Preserves reconstructable rationale for incident prioritization decisions

G-IR-003 — Incident Triage and Routing

Technique	Approach	Relationship	Engineering Rationale
Governed Incident Coordination	Human-Governed Operations	equivalent	Coordinates escalation and routing of operational incidents
Response Workflow Orchestration	Runtime Governance Mediation	strongly-supports	Enables deterministic coordination of incident-response activities
Accountability-Aware Escalation	Accountability Preservation	supports	Preserves attributable governance continuity during incident workflows

G-CONT-001 — Incident Containment Execution

Technique	Approach	Relationship	Engineering Rationale
Operational Containment Engineering	Survivability Engineering	equivalent	Restricts operational propagation during compromise conditions
Runtime Isolation Governance	Trusted Operational Control	strongly-supports	Applies governed operational containment during incident response
Constraint-Aware Mitigation	Safe Operational Behavior	supports	Prevents unsafe or excessive containment activity

G-CONT-002 — Dynamic Isolation and Segmentation

Technique	Approach	Relationship	Engineering Rationale
Adaptive Segmentation Governance	Resilient Operational Control	equivalent	Dynamically adjusts operational isolation boundaries during incidents
Runtime Trust-Boundary Enforcement	Trusted Operational State	strongly-supports	Preserves trusted operational separation under evolving compromise conditions
Survivable Operational Isolation	Survivability Engineering	supports	Limits operational disruption propagation during active incidents

G-CONT-003 — Containment Scope Governance

Technique	Approach	Relationship	Engineering Rationale
Bounded Response Governance	Safe Operational Behavior	equivalent	Restricts containment operations to governed operational scope

Technique	Approach	Relationship	Engineering Rationale
Runtime Constraint Preservation	Deterministic Governance	strongly-supports	Prevents unsafe or uncontrolled remediation activity
Trusted Operational Recovery Coordination	Resilient Operational Control	supports	Maintains continuity while containment operations are executed

G-CONT-004 — Emergency Containment Override

Technique	Approach	Relationship	Engineering Rationale
Emergency Governance Override	Human-Governed Operations	equivalent	Enables authoritative interruption of unsafe operational conditions
Crisis Response Coordination	Survivability Engineering	strongly-supports	Supports rapid mitigation during high-severity operational compromise
Deterministic Emergency Control	Runtime Governance Mediation	supports	Preserves governed emergency response authority

G-FOR-001 — Forensic Evidence Collection

Technique	Approach	Relationship	Engineering Rationale
Operational Forensic Preservation	Evidence-Centric Operations	equivalent	Preserves attributable evidence during incident investigation
Trusted Evidence Integrity	Trusted Operational Assurance	strongly-supports	Maintains integrity and reconstructability of operational evidence
Explainable Incident Reconstruction	Explainable Operational Behavior	supports	Enables forensic reconstruction of operational compromise conditions

G-FOR-002 — Chain-of-Custody Preservation

Technique	Approach	Relationship	Engineering Rationale
Evidence Lineage Preservation	Accountability Preservation	equivalent	Maintains attributable operational custody continuity for forensic evidence
Trusted Evidence Handling	Trusted Operational State	strongly-supports	Prevents tampering or corruption of investigative evidence
Reconstructable Investigative Continuity	Evidence-Centric Operations	supports	Enables verifiable forensic reconstruction workflows

G-FOR-003 — Forensic State Reconstruction

Technique	Approach	Relationship	Engineering Rationale
Operational State Reconstruction	Explainable Operational Behavior	equivalent	Reconstructs operational conditions surrounding incident events
Cross-Domain Evidence Correlation	Trusted Operational Awareness	strongly-supports	Correlates telemetry and operational evidence across governance domains
Trusted Historical Analysis	Trusted Operational Assurance	supports	Enables trustworthy analysis of historical operational state

G-RCA-001 — Root Cause Identification

Technique	Approach	Relationship	Engineering Rationale
Systemic Failure Analysis	Survivability Engineering	equivalent	Determines underlying operational causes of incidents and failures

Technique	Approach	Relationship	Engineering Rationale
Governance-Aware Operational Diagnosis	Adaptive Governance	strongly-supports	Enables governance improvement based on operational-failure analysis
Explainable Failure Attribution	Explainable Operational Behavior	supports	Preserves attributable reasoning for root-cause conclusions

G-RCA-002 — Incident Correlation and Timeline Reconstruction

Technique	Approach	Relationship	Engineering Rationale
Operational Timeline Reconstruction	Evidence-Centric Operations	equivalent	Reconstructs chronological operational activity associated with incidents
Cross-Domain Event Correlation	Trusted Operational Awareness	strongly-supports	Links related operational events across governance domains
Explainable Operational History	Explainable Operational Behavior	supports	Enables understandable reconstruction of incident progression

G-RCA-003 — Corrective Action Coordination

Technique	Approach	Relationship	Engineering Rationale
Corrective Governance Coordination	Adaptive Governance	equivalent	Coordinates operational remediation and governance improvement activities
Recovery-Aware Mitigation Engineering	Resilient Operational Control	strongly-supports	Preserves trusted operational continuity during corrective action workflows
Governed Operational Recovery	Trusted Operational Control	supports	Applies deterministic governance during remediation execution

G-RCA-004 — Lessons Learned and Feedback Integration

Technique	Approach	Relationship	Engineering Rationale
Continuous Governance Adaptation	Adaptive Governance	equivalent	Integrates operational lessons into evolving governance controls
Survivability Improvement Engineering	Survivability Engineering	strongly-supports	Improves enterprise resilience using operational incident findings
Explainable Governance Evolution	Explainable Operational Behavior	supports	Preserves reconstructable governance-improvement rationale

G-EVID-001 — Incident Evidence Generation

Technique	Approach	Relationship	Engineering Rationale
Incident Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence for incident and response workflows
Trusted Operational Accountability	Accountability Preservation	strongly-supports	Maintains reconstructable response-governance continuity
Explainable Incident Governance	Explainable Operational Behavior	supports	Enables explainable review of operational incident handling

G-EVID-002 — Incident Reporting and Documentation

Technique	Approach	Relationship	Engineering Rationale
Structured Incident Documentation	Explainable Operational Behavior	equivalent	Produces structured operational incident records

Technique	Approach	Relationship	Engineering Rationale
Governance-Aware Reporting	Accountability Preservation	strongly-supports	Preserves attributable operational reporting continuity
Operational Transparency Preservation	Trusted Operational Awareness	supports	Enables explainable operational incident communication

G-EVID-003 — Response Attribution and Accountability

Technique	Approach	Relationship	Engineering Rationale
Response Accountability Preservation	Accountability Preservation	equivalent	Maintains attributable operational response history
Action-to-Identity Traceability	Evidence-Centric Operations	strongly-supports	Preserves reconstructable linkage between actors and response activities
Trusted Operational Attribution	Trusted Operational Assurance	supports	Enables trustworthy operational accountability analysis

12. PBSAI Domain H Controls → NIST SP 800-160 Volume 2

H-RES-001 — Operational Resilience Planning

Technique	Approach	Relationship	Engineering Rationale
Resilience-Oriented Systems Engineering	Survivability Engineering	equivalent	Establishes governance structures for resilient operational continuity
Adaptive Continuity Governance	Resilient Operational Control	strongly-supports	Enables operational adaptation during disruption conditions

Technique	Approach	Relationship	Engineering Rationale
Trusted Operational Recovery Coordination	Trusted Operational Control	supports	Preserves governed continuity during adverse operational events

H-RES-002 — Critical Function Prioritization

Technique	Approach	Relationship	Engineering Rationale
Mission-Critical Capability Protection	Trusted Operational Awareness	equivalent	Identifies and prioritizes operationally critical enterprise functions
Dependency-Aware Continuity Engineering	Survivability Engineering	strongly-supports	Preserves continuity of high-impact operational capabilities
Risk-Based Operational Prioritization	Adaptive Governance	supports	Enables continuity decisions according to operational impact severity

H-RES-003 — Resilience Dependency Mapping

Technique	Approach	Relationship	Engineering Rationale
Dependency-Aware Systems Engineering	Trusted Operational Awareness	equivalent	Maps operational dependencies affecting resilience and continuity posture
Cross-Domain Survivability Coordination	Survivability Engineering	strongly-supports	Enables continuity planning across dependent operational domains
Explainable Dependency Governance	Explainable Operational Behavior	supports	Preserves reconstructable understanding of resilience dependencies

H-BKP-001 — Backup Definition and Governance

Technique	Approach	Relationship	Engineering Rationale
Trusted Recovery Preservation	Trusted Operational State	equivalent	Governs recoverable operational backup preservation
Resilient State Continuity	Survivability Engineering	strongly-supports	Maintains recoverable operational state during disruption conditions
Controlled Recovery Governance	Trusted Operational Control	supports	Preserves governed operational recovery authority

H-BKP-002 — Backup Integrity Validation

Technique	Approach	Relationship	Engineering Rationale
Trusted Recovery Integrity Assurance	Trusted Operational Assurance	equivalent	Validates integrity and usability of operational backup artifacts
Recovery Trust Preservation	Trusted Operational State	strongly-supports	Prevents restoration using corrupted or invalid backup data
Resilience-Aware Validation	Survivability Engineering	supports	Maintains dependable recovery readiness conditions

H-BKP-003 — Immutable Backup Protection

Technique	Approach	Relationship	Engineering Rationale
Immutable Recovery Preservation	Survivability Engineering	equivalent	Prevents modification or destruction of critical recovery assets
Trusted Recovery State Protection	Trusted Operational State	strongly-supports	Maintains trustworthy backup continuity under adversarial conditions

Technique	Approach	Relationship	Engineering Rationale
Compromise-Resistant Continuity Engineering	Resilient Operational Control	supports	Reduces operational exposure during destructive compromise scenarios

H-REC-001 — Recovery Planning

Technique	Approach	Relationship	Engineering Rationale
Recovery-Oriented Systems Engineering	Survivability Engineering	equivalent	Defines structured operational recovery and restoration workflows
Trusted Recovery Coordination	Trusted Operational Control	strongly-supports	Preserves deterministic governance during operational restoration
Explainable Recovery Governance	Explainable Operational Behavior	supports	Enables reconstructable analysis of recovery planning rationale

H-REC-002 — Recovery Dependency Coordination

Technique	Approach	Relationship	Engineering Rationale
Dependency-Aware Recovery Engineering	Trusted Operational Awareness	equivalent	Coordinates restoration according to operational dependency relationships
Cross-Domain Recovery Governance	Resilient Operational Control	strongly-supports	Preserves continuity across interdependent enterprise systems
Controlled Restoration Sequencing	Trusted Operational State	supports	Maintains orderly recovery of trusted operational state

H-REC-003 — Recovery Execution

Technique	Approach	Relationship	Engineering Rationale
Controlled Operational Restoration	Trusted Operational Control	equivalent	Executes governed recovery and restoration workflows
Runtime Recovery Governance	Runtime Governance Mediation	strongly-supports	Preserves deterministic governance during operational restoration
Trusted State Reconstitution	Trusted Operational State	supports	Restores valid and governed operational execution conditions

H-REC-004 — Recovery Validation and Assurance

Technique	Approach	Relationship	Engineering Rationale
Recovery Assurance Engineering	Trusted Operational Assurance	equivalent	Validates correctness and trustworthiness of restored operational state
Resilient Operational Validation	Survivability Engineering	strongly-supports	Ensures dependable recovery prior to resumption of operations
Explainable Restoration Verification	Explainable Operational Behavior	supports	Enables reconstructable recovery-validation analysis

H-REC-005 — Recovery Authorization Governance

Technique	Approach	Relationship	Engineering Rationale
Authorized Recovery Governance	Human-Governed Operations	equivalent	Requires governed authorization prior to restoration activities
Deterministic Recovery Mediation	Runtime Governance Mediation	strongly-supports	Prevents unauthorized or unsafe operational restoration

Technique	Approach	Relationship	Engineering Rationale
Accountability-Aware Recovery Control	Accountability Preservation	supports	Preserves attributable governance continuity during recovery workflows

H-REC-006 — Recovery Rollback and Safe Restoration

Technique	Approach	Relationship	Engineering Rationale
Safe Restoration Engineering	Safe Operational Behavior	equivalent	Enables rollback and controlled restoration following recovery failure
Trusted Recovery Reversal	Trusted Operational State	strongly-supports	Prevents persistence of invalid restored operational conditions
Resilience-Oriented Restoration Governance	Survivability Engineering	supports	Preserves continuity during restoration correction workflows

H-FAIL-001 — Failover Orchestration

Technique	Approach	Relationship	Engineering Rationale
Continuity-Oriented Failover Engineering	Survivability Engineering	equivalent	Maintains operational continuity during service disruption conditions
Runtime Continuity Coordination	Resilient Operational Control	strongly-supports	Preserves governed failover execution under operational stress
Trusted Operational Transition	Trusted Operational State	supports	Maintains continuity of trusted operational execution state

H-FAIL-002 — Redundant Capability Coordination

Technique	Approach	Relationship	Engineering Rationale
Redundancy-Aware Systems Engineering	Survivability Engineering	equivalent	Coordinates redundant operational capabilities for continuity assurance
Operational Availability Preservation	Trusted Operational Assurance	strongly-supports	Maintains dependable operational service continuity
Dependency-Aware Redundancy Governance	Trusted Operational Awareness	supports	Preserves visibility into continuity-supporting operational dependencies

H-TEST-001 — Continuity and Recovery Testing

Technique	Approach	Relationship	Engineering Rationale
Resilience Validation Engineering	Trusted Operational Assurance	equivalent	Validates operational continuity and recovery readiness
Survivability Readiness Assessment	Survivability Engineering	strongly-supports	Evaluates operational resilience under disruption conditions
Explainable Resilience Verification	Explainable Operational Behavior	supports	Preserves reconstructable resilience-testing rationale and evidence

H-TEST-002 — Recovery Scenario Simulation

Technique	Approach	Relationship	Engineering Rationale
Adversarial Recovery Simulation	Survivability Engineering	equivalent	Simulates operational disruption and recovery conditions
Adaptive Continuity Evaluation	Adaptive Governance	strongly-supports	Enables governance refinement through simulated operational stress conditions

Technique	Approach	Relationship	Engineering Rationale
Runtime Recovery Observation	Trusted Operational Awareness	supports	Preserves operational visibility during resilience exercises

H-TEST-003 — Tabletop and Governance Exercises

Technique	Approach	Relationship	Engineering Rationale
Governance Readiness Engineering	Human-Governed Operations	equivalent	Evaluates governance decision-making during continuity scenarios
Operational Coordination Validation	Trusted Operational Control	strongly-supports	Validates coordinated response and continuity governance workflows
Explainable Governance Evaluation	Explainable Operational Behavior	supports	Enables reconstructable review of operational exercise outcomes

H-TEST-004 — Recovery Metrics and Performance Evaluation

Technique	Approach	Relationship	Engineering Rationale
Operational Resilience Analytics	Trusted Operational Assurance	equivalent	Measures continuity and recovery effectiveness
Adaptive Recovery Optimization	Adaptive Governance	strongly-supports	Enables governance improvement using resilience-performance metrics
Evidence-Aware Resilience Evaluation	Evidence-Centric Operations	supports	Preserves attributable continuity-performance evidence

H-TEST-005 — Improvement and Feedback Integration

Technique	Approach	Relationship	Engineering Rationale
Adaptive Resilience Governance	Adaptive Governance	equivalent	Integrates resilience findings into evolving governance controls
Survivability Improvement Engineering	Survivability Engineering	strongly-supports	Continuously improves operational continuity capabilities
Explainable Governance Evolution	Explainable Operational Behavior	supports	Preserves reconstructable rationale for resilience-governance adaptation

H-EVID-001 — Resilience Event Logging

Technique	Approach	Relationship	Engineering Rationale
Continuity Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable resilience and continuity telemetry
Accountability-Aware Recovery Governance	Accountability Preservation	strongly-supports	Preserves reconstructable continuity-governance history
Explainable Recovery Reconstruction	Explainable Operational Behavior	supports	Enables explainable analysis of resilience and restoration workflows

H-EVID-002 — Recovery State Traceability

Technique	Approach	Relationship	Engineering Rationale
Recovery State Lineage Preservation	Trusted Operational State	equivalent	Maintains reconstructable continuity of operational recovery state
Restoration Traceability Engineering	Evidence-Centric Operations	strongly-supports	Enables attributable analysis of restoration progression

Technique	Approach	Relationship	Engineering Rationale
Trusted Recovery Visibility	Trusted Operational Awareness	supports	Preserves operational visibility into recovery-state transitions

13. PBSAI Domain I Controls → NIST SP 800-160 Volume 2

I-ARCH-001 — Security Architecture Definition

Technique	Approach	Relationship	Engineering Rationale
Secure Architecture Engineering	Trusted Operational Control	equivalent	Establishes authoritative secure architecture patterns for governed enterprise systems
Defense-in-Depth Design	Survivability Engineering	strongly-supports	Reduces operational fragility through layered protective architecture
Explainable Architecture Governance	Explainable Operational Behavior	supports	Enables reconstructable review of architectural governance decisions

I-ARCH-002 — Architecture Review and Validation

Technique	Approach	Relationship	Engineering Rationale
Architecture Assurance Engineering	Trusted Operational Assurance	equivalent	Validates system architecture against security and governance expectations
Design Integrity Validation	Trusted Operational State	strongly-supports	Prevents introduction of untrusted or misaligned architectural patterns
Evidence-Aware Architecture Review	Evidence-Centric Operations	supports	Preserves attributable evidence of architectural validation decisions

I-ARCH-003 — Trust Boundary Definition

Technique	Approach	Relationship	Engineering Rationale
Trust Boundary Engineering	Trusted Operational State	equivalent	Defines operational trust boundaries for secure system interaction
Boundary Protection Design	Survivability Engineering	strongly-supports	Reduces propagation of compromise across operational domains
Context-Aware Boundary Governance	Runtime Governance Mediation	supports	Enables governance decisions based on trust-boundary context

I-ARCH-004 — Architecture Exception Governance

Technique	Approach	Relationship	Engineering Rationale
Exception Governance Engineering	Adaptive Governance	equivalent	Applies governed review to deviations from approved architecture
Risk-Aware Architecture Control	Trusted Operational Control	strongly-supports	Prevents unmanaged architectural exceptions from weakening system trustworthiness
Explainable Exception Traceability	Explainable Operational Behavior	supports	Preserves reconstructable rationale for approved architecture exceptions

I-ARCH-005 — Architecture Traceability and Evidence

Technique	Approach	Relationship	Engineering Rationale
Architecture Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence for architecture decisions and validation
Design Traceability Engineering	Accountability Preservation	strongly-supports	Preserves lineage between architecture, implementation, and operational governance

Technique	Approach	Relationship	Engineering Rationale
Explainable Design Reconstruction	Explainable Operational Behavior	supports	Enables reconstruction of system design and security-governance rationale

I-SEG-001 — Security Segmentation Design

Technique	Approach	Relationship	Engineering Rationale
Segmentation Architecture Engineering	Survivability Engineering	equivalent	Designs segmentation to limit compromise propagation and operational exposure
Trust-Boundary Partitioning	Trusted Operational State	strongly-supports	Establishes partitioned operational trust domains
Boundary-Aware Governance	Runtime Governance Mediation	supports	Supports governance decisions based on segmented operational context

I-SEG-002 — Inter-Domain Communication Governance

Technique	Approach	Relationship	Engineering Rationale
Controlled Information Flow Engineering	Trusted Operational Control	equivalent	Governs communication between operational domains and trust zones
Cross-Domain Constraint Enforcement	Safe Operational Behavior	strongly-supports	Prevents unauthorized or unsafe inter-domain interaction
Explainable Flow Governance	Explainable Operational Behavior	supports	Enables review and reconstruction of permitted communication pathways

I-SEG-003 — Isolation Verification

Technique	Approach	Relationship	Engineering Rationale
Isolation Assurance Engineering	Trusted Operational Assurance	equivalent	Validates effectiveness of operational isolation and segmentation
Boundary Integrity Monitoring	Trusted Operational State	strongly-supports	Detects segmentation drift or boundary-control degradation
Evidence-Aware Isolation Validation	Evidence-Centric Operations	supports	Preserves attributable evidence of boundary-assurance outcomes

I-SEG-004 — Dynamic Segmentation Enforcement

Technique	Approach	Relationship	Engineering Rationale
Adaptive Segmentation Control	Resilient Operational Control	equivalent	Dynamically adjusts segmentation boundaries during evolving operational conditions
Runtime Boundary Enforcement	Trusted Operational State	strongly-supports	Maintains operational separation under changing threat or risk conditions
Survivable Isolation Governance	Survivability Engineering	supports	Limits compromise propagation through adaptive containment boundaries

I-SEG-005 — Segmentation Auditability

Technique	Approach	Relationship	Engineering Rationale
Boundary-Control Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence for segmentation and boundary changes
Segmentation Accountability Engineering	Accountability Preservation	strongly-supports	Preserves reconstructable operational boundary-governance history

Technique	Approach	Relationship	Engineering Rationale
Explainable Boundary Reconstruction	Explainable Operational Behavior	supports	Enables review of segmentation state and governance decisions

I-INF-001 — Infrastructure Hardening Baselines

Technique	Approach	Relationship	Engineering Rationale
Secure Baseline Engineering	Trusted Operational State	equivalent	Establishes hardened infrastructure states as trusted operational baselines
Least Functionality Engineering	Safe Operational Behavior	strongly-supports	Reduces attack surface through minimized unnecessary capability
Baseline Governance Traceability	Evidence-Centric Operations	supports	Preserves evidence of hardening decisions and baseline adoption

I-INF-002 — Hardening Validation

Technique	Approach	Relationship	Engineering Rationale
Baseline Compliance Assurance	Trusted Operational Assurance	equivalent	Validates infrastructure conformance to hardened baseline requirements
Operational Integrity Verification	Trusted Operational State	strongly-supports	Prevents operation under weakened or nonconforming configurations
Evidence-Aware Hardening Review	Evidence-Centric Operations	supports	Preserves attributable evidence of hardening validation outcomes

I-INF-003 — Unauthorized Configuration Detection

Technique	Approach	Relationship	Engineering Rationale
Configuration Integrity Monitoring	Trusted Operational Assurance	equivalent	Detects unauthorized modification to hardened infrastructure state
Drift Detection Engineering	Adaptive Governance	strongly-supports	Enables governance response to nonconforming operational configuration
Trusted State Preservation	Trusted Operational State	supports	Maintains visibility into deviations from trusted operating conditions

I-INF-004 — Secure Platform Provisioning

Technique	Approach	Relationship	Engineering Rationale
Trusted Platform Provisioning	Trusted Operational Control	equivalent	Ensures platforms are provisioned according to governed secure baselines
Runtime Deployment Governance	Runtime Governance Mediation	strongly-supports	Applies governance before operational activation of platform resources
Secure State Initialization	Trusted Operational State	supports	Establishes trusted operating conditions at provisioning time

I-INF-005 — Infrastructure Governance Evidence

Technique	Approach	Relationship	Engineering Rationale
Infrastructure Evidence Preservation	Evidence-Centric Operations	equivalent	Produces evidence for hardening, validation, and platform governance
Operational Accountability Preservation	Accountability Preservation	strongly-supports	Maintains reconstructable infrastructure governance history

Technique	Approach	Relationship	Engineering Rationale
Explainable Platform Governance	Explainable Operational Behavior	supports	Enables review of infrastructure security decisions and outcomes

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I-DEP-001 — Deployment Architecture Governance

Technique	Approach	Relationship	Engineering Rationale
Trusted Deployment Engineering	Trusted Operational Control	equivalent	Governs deployment architecture according to approved operational design constraints
Runtime Activation Governance	Runtime Governance Mediation	strongly-supports	Prevents unauthorized or unsafe deployment activation
Controlled Operational Transition	Trusted Operational State	supports	Preserves trusted execution conditions during deployment lifecycle transitions

I-DEP-002 — Environment Separation Governance

Technique	Approach	Relationship	Engineering Rationale
Environment Isolation Engineering	Survivability Engineering	equivalent	Separates development, testing, staging, and production operational domains
Trust-Boundary Segregation	Trusted Operational State	strongly-supports	Prevents unsafe operational crossover between environments
Controlled Deployment Governance	Trusted Operational Control	supports	Reduces exposure arising from improperly governed deployment workflows

I-DEP-003 — Deployment Scope Constraints

Technique	Approach	Relationship	Engineering Rationale
Bounded Deployment Governance	Safe Operational Behavior	equivalent	Restricts deployment authority and operational execution scope
Constraint-Aware Release Management	Runtime Governance Mediation	strongly-supports	Prevents unsafe or unauthorized deployment propagation
Trusted Operational Limitation	Trusted Operational Control	supports	Maintains governance control over deployment reach and impact

I-DEP-004 — Secure Deployment Validation

Technique	Approach	Relationship	Engineering Rationale
Deployment Assurance Engineering	Trusted Operational Assurance	equivalent	Validates security and governance posture prior to operational activation
Runtime Integrity Verification	Trusted Operational State	strongly-supports	Prevents deployment of untrusted or nonconforming operational artifacts
Evidence-Aware Release Governance	Evidence-Centric Operations	supports	Preserves attributable deployment validation evidence

I-DEP-005 — Deployment Rollback Governance

Technique	Approach	Relationship	Engineering Rationale
Safe Deployment Reversal	Survivability Engineering	equivalent	Enables controlled rollback of failed or unsafe operational deployments
Trusted Restoration Governance	Resilient Operational Control	strongly-supports	Preserves trusted operational continuity during rollback workflows

Technique	Approach	Relationship	Engineering Rationale
Runtime Recovery Coordination	Runtime Governance Mediation	supports	Enables governed restoration of operational deployment state

I-SSE-001 — Secure Systems Engineering Governance

Technique	Approach	Relationship	Engineering Rationale
Secure Systems Engineering	Trusted Operational Control	equivalent	Establishes governance-oriented systems-security engineering practices
Governance-Aware Design Assurance	Trusted Operational Assurance	strongly-supports	Aligns operational systems engineering with governance and trust objectives
Explainable Engineering Traceability	Explainable Operational Behavior	supports	Preserves reconstructable rationale for secure engineering decisions

I-SSE-002 — Threat Modeling and Design Analysis

Technique	Approach	Relationship	Engineering Rationale
Threat-Informed Systems Engineering	Survivability Engineering	equivalent	Models operational attack paths and trust degradation conditions
Adversarial Exposure Analysis	Adaptive Operational Awareness	strongly-supports	Identifies operational weaknesses and architectural exposure pathways
Explainable Threat Reasoning	Explainable Operational Behavior	supports	Enables reconstructable analysis of threat-modeling assumptions and outcomes

I-SSE-003 — Security-by-Design Enforcement

Technique	Approach	Relationship	Engineering Rationale
Secure-by-Default Engineering	Safe Operational Behavior	equivalent	Embeds governance and security controls directly into operational design
Deterministic Security Mediation	Runtime Governance Mediation	strongly-supports	Prevents unsafe operational behavior through built-in governance enforcement
Trusted Architectural Foundations	Trusted Operational State	supports	Establishes trustworthy operational design baselines

I-SSE-004 — Secure Interface and API Governance

Technique	Approach	Relationship	Engineering Rationale
Trusted Interface Engineering	Trusted Operational Interaction	equivalent	Governs operational APIs and inter-system interaction pathways
Runtime Interaction Validation	Trusted Operational Control	strongly-supports	Prevents unsafe or unauthorized operational interaction
Explainable Interface Governance	Explainable Operational Behavior	supports	Enables reconstructable analysis of governed interaction decisions

I-SSE-005 — Engineering Evidence and Design Traceability

Technique	Approach	Relationship	Engineering Rationale
Systems Engineering Traceability	Evidence-Centric Operations	equivalent	Preserves attributable lineage between architecture, engineering, and operations
Accountability-Aware Design Governance	Accountability Preservation	strongly-supports	Maintains reconstructable systems-engineering governance continuity

Technique	Approach	Relationship	Engineering Rationale
Explainable Engineering Reconstruction	Explainable Operational Behavior	supports	Enables review and analysis of engineering lifecycle decisions

I-ASSUR-001 — Architecture Assurance Validation

Technique	Approach	Relationship	Engineering Rationale
Architecture Assurance Engineering	Trusted Operational Assurance	equivalent	Validates trustworthiness of operational architecture implementation
Trusted Design Verification	Trusted Operational State	strongly-supports	Ensures operational systems conform to governed architectural expectations
Assurance-Aware Governance	Adaptive Governance	supports	Enables governance refinement based on assurance findings

I-ASSUR-002 — Cross-Domain Dependency Validation

Technique	Approach	Relationship	Engineering Rationale
Dependency Assurance Engineering	Trusted Operational Awareness	equivalent	Validates operational trust relationships across dependent systems and domains
Cross-Domain Survivability Analysis	Survivability Engineering	strongly-supports	Identifies resilience exposure introduced by operational dependencies
Runtime Dependency Governance	Runtime Governance Mediation	supports	Preserves governance visibility into evolving dependency relationships

I-ASSUR-003 — Continuous Architectural Monitoring

Technique	Approach	Relationship	Engineering Rationale
Continuous Architecture Assurance	Trusted Operational Assurance	equivalent	Continuously evaluates operational architecture integrity and conformance
Runtime Drift Detection	Adaptive Governance	strongly-supports	Detects deviation from governed architectural expectations
Trusted Operational Visibility	Trusted Operational Awareness	supports	Maintains awareness of architectural trustworthiness conditions

I-ASSUR-004 — Assurance Escalation Governance

Technique	Approach	Relationship	Engineering Rationale
Assurance Escalation Engineering	Human-Governed Operations	equivalent	Escalates high-impact assurance findings for governed operational review
Risk-Aware Governance Coordination	Adaptive Governance	strongly-supports	Coordinates governance response to assurance degradation conditions
Accountability-Aware Escalation	Accountability Preservation	supports	Preserves attributable assurance-escalation history

I-ASSUR-005 — Architecture Assurance Evidence

Technique	Approach	Relationship	Engineering Rationale
Assurance Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence for architecture and assurance validation activities
Trusted Assurance Traceability	Accountability Preservation	strongly-supports	Maintains reconstructable lineage for assurance-related governance decisions

Technique	Approach	Relationship	Engineering Rationale
Explainable Assurance Reconstruction	Explainable Operational Behavior	supports	Enables review and interpretation of architecture assurance outcomes

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J-PHYS-001 — Physical Security Governance

Technique	Approach	Relationship	Engineering Rationale
Physical Protection Engineering	Trusted Operational Control	equivalent	Establishes governance for protection of physical operational infrastructure
Facility Access Constraint Enforcement	Safe Operational Behavior	strongly-supports	Restricts unauthorized physical interaction with governed systems
Trusted Facility Governance	Trusted Operational State	supports	Preserves trusted environmental operating conditions

J-PHYS-002 — Facility Access Authorization

Technique	Approach	Relationship	Engineering Rationale
Authorized Physical Access Control	Human-Governed Operations	equivalent	Validates physical access authority prior to facility entry
Identity-Aware Facility Governance	Trusted Operational Interaction	strongly-supports	Associates facility access with governed operational identity
Accountability-Aware Entry Traceability	Accountability Preservation	supports	Maintains attributable facility-access history

J-PHYS-003 — Secure Facility Segmentation

Technique	Approach	Relationship	Engineering Rationale
Physical Segmentation Engineering	Survivability Engineering	equivalent	Separates operational environments according to security and trust requirements
Trust-Boundary Facility Protection	Trusted Operational State	strongly-supports	Preserves isolation between sensitive operational domains
Controlled Physical Interaction	Safe Operational Behavior	supports	Limits exposure arising from unrestricted facility access

J-PHYS-004 — Environmental Condition Monitoring

Technique	Approach	Relationship	Engineering Rationale
Environmental Operational Awareness	Trusted Operational Awareness	equivalent	Monitors environmental conditions affecting operational trustworthiness
Runtime Environmental Assurance	Trusted Operational Assurance	strongly-supports	Detects unsafe environmental conditions impacting system reliability
Adaptive Environmental Governance	Adaptive Governance	supports	Enables operational adjustment during changing environmental conditions

J-PHYS-005 — Physical Security Event Logging

Technique	Approach	Relationship	Engineering Rationale
Physical Event Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence for facility-security operations
Physical Accountability Preservation	Accountability Preservation	strongly-supports	Maintains reconstructable operational facility-governance lineage

Technique	Approach	Relationship	Engineering Rationale
Explainable Physical Security Reconstruction	Explainable Operational Behavior	supports	Enables review of facility-security operational activity

J-SURV-001 — Surveillance and Monitoring Governance

Technique	Approach	Relationship	Engineering Rationale
Operational Surveillance Engineering	Trusted Operational Awareness	equivalent	Establishes governed operational surveillance and monitoring capabilities
Continuous Physical Observation	Trusted Operational Assurance	strongly-supports	Maintains awareness of physical operational conditions
Governance-Aware Monitoring	Runtime Governance Mediation	supports	Preserves controlled operational use of surveillance systems

J-SURV-002 — Physical Anomaly Detection

Technique	Approach	Relationship	Engineering Rationale
Physical Threat Detection	Survivability Engineering	equivalent	Detects anomalous physical operational conditions and threat indicators
Runtime Situational Awareness	Trusted Operational Awareness	strongly-supports	Maintains operational awareness of physical compromise conditions
Adaptive Protective Response	Resilient Operational Control	supports	Enables operational adaptation during detected physical anomalies

J-SURV-003 — Cross-Domain Event Correlation

Technique	Approach	Relationship	Engineering Rationale
Multi-Domain Situational Awareness	Trusted Operational Awareness	equivalent	Correlates cyber, physical, and environmental operational events
Cross-Domain Survivability Analysis	Survivability Engineering	strongly-supports	Detects compound operational compromise conditions
Explainable Event Correlation	Explainable Operational Behavior	supports	Enables reconstructable understanding of multi-domain operational events

J-SURV-004 — Physical Incident Escalation

Technique	Approach	Relationship	Engineering Rationale
Escalation-Oriented Governance	Human-Governed Operations	equivalent	Coordinates operational escalation for high-impact physical incidents
Crisis-Aware Response Coordination	Survivability Engineering	strongly-supports	Enables rapid response to facility compromise conditions
Accountability-Aware Escalation	Accountability Preservation	supports	Preserves attributable operational escalation history

J-ENV-001 — Environmental Resilience Governance

Technique	Approach	Relationship	Engineering Rationale
Environmental Survivability Engineering	Survivability Engineering	equivalent	Governs resilience against environmental disruption conditions
Adaptive Environmental Continuity	Resilient Operational Control	strongly-supports	Preserves operational continuity during environmental instability

Technique	Approach	Relationship	Engineering Rationale
Trusted Operational Environment Preservation	Trusted Operational State	supports	Maintains trustworthy environmental operating conditions

J-ENV-002 — Power and Utility Continuity

Technique	Approach	Relationship	Engineering Rationale
Utility Continuity Engineering	Survivability Engineering	equivalent	Preserves operational continuity during utility disruption conditions
Redundant Infrastructure Coordination	Resilient Operational Control	strongly-supports	Maintains dependable operational availability
Trusted Infrastructure Awareness	Trusted Operational Awareness	supports	Preserves visibility into utility dependency conditions

J-ENV-003 — Environmental Recovery Coordination

Technique	Approach	Relationship	Engineering Rationale
Environmental Recovery Engineering	Survivability Engineering	equivalent	Coordinates restoration following environmental disruption events
Controlled Environmental Restoration	Trusted Operational Control	strongly-supports	Preserves governed recovery workflows during environmental incidents
Trusted Facility Restoration	Trusted Operational State	supports	Restores valid operational environmental conditions

J-ENV-004 — Environmental Assurance Validation

Technique	Approach	Relationship	Engineering Rationale
Environmental Assurance Engineering	Trusted Operational Assurance	equivalent	Validates environmental resilience and operational safety posture
Runtime Condition Validation	Trusted Operational State	strongly-supports	Ensures systems operate within trusted environmental conditions
Evidence-Aware Environmental Governance	Evidence-Centric Operations	supports	Preserves attributable environmental assurance evidence

J-EVID-001 — Physical and Environmental Evidence Preservation

Technique	Approach	Relationship	Engineering Rationale
Environmental Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable records for physical and environmental operational events
Cross-Domain Accountability Preservation	Accountability Preservation	strongly-supports	Maintains reconstructable physical-environmental governance lineage
Explainable Environmental Reconstruction	Explainable Operational Behavior	supports	Enables review and analysis of physical-environmental operational conditions

15. PBSAI Domain K Controls → NIST SP 800-160 Volume 2

K-SCRM-001 — Supply Chain Risk Assessment

Technique	Approach	Relationship	Engineering Rationale
Supply Chain Survivability Engineering	Survivability Engineering	equivalent	Evaluates operational exposure arising from suppliers, vendors, and external dependencies

Technique	Approach	Relationship	Engineering Rationale
Dependency-Aware Risk Governance	Adaptive Governance	strongly-supports	Enables governance decisions informed by evolving supplier risk posture
Trusted Supplier Visibility	Trusted Operational Awareness	supports	Maintains awareness of supply-chain trust relationships

K-SCRM-002 — Supplier Trust Evaluation

Technique	Approach	Relationship	Engineering Rationale
Trusted Supplier Assessment	Trusted Operational Assurance	equivalent	Validates trustworthiness and governance posture of suppliers and service providers
Dependency Integrity Validation	Trusted Operational State	strongly-supports	Prevents reliance on untrusted or nonconforming supply-chain entities
Explainable Supplier Governance	Explainable Operational Behavior	supports	Preserves reconstructable rationale for supplier trust decisions

K-SCRM-003 — Third-Party Dependency Governance

Technique	Approach	Relationship	Engineering Rationale
Dependency Governance Engineering	Trusted Operational Control	equivalent	Applies governance constraints to third-party operational dependencies
Runtime Dependency Mediation	Runtime Governance Mediation	strongly-supports	Preserves controlled operational interaction with external systems and services
Bounded Dependency Exposure	Safe Operational Behavior	supports	Restricts unsafe operational dependency expansion

K-SCRM-004 — Supply Chain Monitoring

Technique	Approach	Relationship	Engineering Rationale
Continuous Dependency Observation	Trusted Operational Awareness	equivalent	Continuously monitors supplier and dependency operational conditions
Runtime Trust Evaluation	Adaptive Operational Awareness	strongly-supports	Detects evolving supply-chain trust degradation conditions
Survivability-Aware Dependency Monitoring	Survivability Engineering	supports	Improves resilience against supply-chain compromise propagation

K-SCRM-005 — Supply Chain Incident Coordination

Technique	Approach	Relationship	Engineering Rationale
Supply Chain Response Coordination	Resilient Operational Control	equivalent	Coordinates governance response during supply-chain compromise conditions
Cross-Organizational Recovery Governance	Survivability Engineering	strongly-supports	Preserves continuity during dependency-related operational disruption
Accountability-Aware Dependency Response	Accountability Preservation	supports	Maintains attributable supply-chain incident governance history

K-SBOM-001 — SBOM Generation and Maintenance

Technique	Approach	Relationship	Engineering Rationale
Dependency Visibility Engineering	Trusted Operational Awareness	equivalent	Maintains authoritative inventory of software dependencies and components

Technique	Approach	Relationship	Engineering Rationale
Operational Dependency Traceability	Evidence-Centric Operations	strongly-supports	Enables reconstructable analysis of operational dependency lineage
Trusted Component Awareness	Trusted Operational State	supports	Preserves visibility into operational component trust relationships

K-SBOM-002 — Dependency Risk Analysis

Technique	Approach	Relationship	Engineering Rationale
Dependency Exposure Analysis	Survivability Engineering	equivalent	Evaluates operational risk arising from software and infrastructure dependencies
Vulnerability-Aware Governance	Adaptive Governance	strongly-supports	Enables mitigation decisions informed by dependency exposure
Runtime Dependency Visibility	Trusted Operational Awareness	supports	Preserves awareness of evolving operational dependency conditions

K-SBOM-003 — Component Integrity Validation

Technique	Approach	Relationship	Engineering Rationale
Trusted Component Verification	Trusted Operational Assurance	equivalent	Validates integrity and authenticity of operational components
Supply Chain Integrity Preservation	Trusted Operational State	strongly-supports	Prevents operational use of tampered or untrusted components
Evidence-Aware Integrity Governance	Evidence-Centric Operations	supports	Preserves attributable validation evidence for component trustworthiness

K-SBOM-004 — Dependency Change Monitoring

Technique	Approach	Relationship	Engineering Rationale
Dependency Drift Detection	Adaptive Governance	equivalent	Detects operational changes affecting governed dependencies
Runtime Dependency Integrity Monitoring	Trusted Operational Assurance	strongly-supports	Maintains visibility into evolving component trust conditions
Trusted Operational Dependency Continuity	Trusted Operational State	supports	Preserves dependable operational dependency posture

K-SBOM-005 — Dependency Governance Evidence

Technique	Approach	Relationship	Engineering Rationale
Dependency Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable records for dependency governance activities
Supply Chain Accountability Preservation	Accountability Preservation	strongly-supports	Maintains reconstructable operational dependency-governance lineage
Explainable Dependency Reconstruction	Explainable Operational Behavior	supports	Enables review of dependency decisions and trust relationships

K-PROV-001 — Component Provenance Validation

Technique	Approach	Relationship	Engineering Rationale
Provenance Assurance Engineering	Trusted Operational Assurance	equivalent	Validates lineage and origin of operational components

Technique	Approach	Relationship	Engineering Rationale
Trusted Artifact Lineage Preservation	Accountability Preservation	strongly-supports	Maintains reconstructable provenance continuity for operational artifacts
Explainable Provenance Analysis	Explainable Operational Behavior	supports	Enables review of component origin and trustworthiness decisions

K-PROV-002 — Provenance Metadata Preservation

Technique	Approach	Relationship	Engineering Rationale
Provenance Continuity Engineering	Evidence-Centric Operations	equivalent	Preserves provenance metadata throughout operational workflows
Runtime Artifact Traceability	Trusted Operational Awareness	strongly-supports	Enables operational awareness of artifact lineage and transformation
Trusted Metadata Governance	Trusted Operational State	supports	Preserves integrity of provenance metadata across lifecycle operations

K-PROV-003 — Supply Chain Attestation

Technique	Approach	Relationship	Engineering Rationale
Attested Supply Chain Governance	Trusted Operational Assurance	equivalent	Produces verifiable claims regarding supply-chain trustworthiness
Dependency Integrity Validation	Trusted Operational State	strongly-supports	Ensures operational dependencies satisfy governed trust expectations
Explainable Trust Verification	Explainable Operational Behavior	supports	Enables review and interpretation of supply-chain attestation evidence

K-PROV-004 — Artifact Signing and Verification

Technique	Approach	Relationship	Engineering Rationale
Trusted Artifact Authentication	Trusted Operational Control	equivalent	Validates authenticity of operational software and deployment artifacts
Integrity-Aware Distribution Governance	Survivability Engineering	strongly-supports	Prevents operational use of unauthorized or modified artifacts
Runtime Trust Verification	Trusted Operational Assurance	supports	Maintains confidence in distributed operational components

K-LC-001 — Secure Lifecycle Governance

Technique	Approach	Relationship	Engineering Rationale
Lifecycle Security Engineering	Trusted Operational Control	equivalent	Governs secure operational lifecycle activities across enterprise systems
Continuous Lifecycle Assurance	Trusted Operational Assurance	strongly-supports	Maintains trustworthiness throughout system lifecycle transitions
Explainable Lifecycle Governance	Explainable Operational Behavior	supports	Preserves reconstructable lifecycle governance rationale

K-LC-002 — Secure Development Governance

Technique	Approach	Relationship	Engineering Rationale
Secure Development Engineering	Safe Operational Behavior	equivalent	Applies secure-by-design governance during development activities

Technique	Approach	Relationship	Engineering Rationale
Governance-Aware Engineering Control	Runtime Governance Mediation	strongly-supports	Preserves governed engineering workflows and operational discipline
Trusted Development Continuity	Trusted Operational State	supports	Maintains trusted engineering and development conditions

K-LC-003 — Lifecycle Integrity Validation

Technique	Approach	Relationship	Engineering Rationale
Lifecycle Integrity Assurance	Trusted Operational Assurance	equivalent	Validates integrity of operational systems throughout lifecycle progression
Continuous Trust Evaluation	Adaptive Governance	strongly-supports	Detects trust degradation during lifecycle evolution
Accountability-Aware Lifecycle Evidence	Accountability Preservation	supports	Preserves reconstructable lifecycle validation history

K-LC-004 — End-of-Life Governance

Technique	Approach	Relationship	Engineering Rationale
Controlled Lifecycle Termination	Trusted Operational Control	equivalent	Governs secure retirement and decommissioning of operational systems
Residual Exposure Reduction	Survivability Engineering	strongly-supports	Prevents persistence of unmanaged operational risk after retirement
Explainable Lifecycle Closure	Explainable Operational Behavior	supports	Enables reconstructable end-of-life governance review

K-EVID-001 — Supply Chain and Lifecycle Evidence Preservation

Technique	Approach	Relationship	Engineering Rationale
Lifecycle Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence for supply-chain and lifecycle governance activities
Dependency Accountability Preservation	Accountability Preservation	strongly-supports	Maintains reconstructable operational lineage across lifecycle operations
Explainable Supply Chain Reconstruction	Explainable Operational Behavior	supports	Enables review and interpretation of supply-chain governance outcomes

16. PBSAI Domain L Controls → NIST SP 800-160 Volume 2

L-PROG-001 — Security Program Governance

Technique	Approach	Relationship	Engineering Rationale
Enterprise Security Governance	Governance Accountability	equivalent	Establishes program-level governance for secure AI-enabled enterprise operations
Coordinated Security Engineering	Trusted Operational Control	strongly-supports	Aligns security engineering responsibilities across operational domains
Evidence-Aware Program Oversight	Evidence-Centric Operations	supports	Preserves attributable security-program governance evidence

L-PROG-002 — Program Risk Integration

Technique	Approach	Relationship	Engineering Rationale
Risk-Informed Program Governance	Adaptive Governance	equivalent	Integrates operational risk into enterprise security-program decisions
Cross-Domain Risk Coordination	Resilient Operational Awareness	strongly-supports	Correlates risk conditions across operational and governance domains

Technique	Approach	Relationship	Engineering Rationale
Explainable Risk Governance	Explainable Operational Behavior	supports	Preserves reconstructable rationale for program-level risk decisions

L-PROG-003 — Security Program Metrics

Technique	Approach	Relationship	Engineering Rationale
Operational Governance Measurement	Trusted Operational Assurance	equivalent	Measures security-program performance and governance effectiveness
Continuous Assurance Analytics	Adaptive Governance	strongly-supports	Enables improvement of operational governance through measured outcomes
Accountability-Aware Metrics	Evidence-Centric Operations	supports	Preserves attributable evidence for program effectiveness analysis

L-PROG-004 — Governance Reporting and Escalation

Technique	Approach	Relationship	Engineering Rationale
Governance Reporting Engineering	Accountability Preservation	equivalent	Supports accountable reporting and escalation of governance conditions
Supervisory Escalation Governance	Human-Governed Operations	strongly-supports	Enables human review for significant governance or operational risk conditions
Explainable Governance Visibility	Explainable Operational Behavior	supports	Preserves visibility into governance status and escalation rationale

L-AI-001 — AI Model Governance

Technique	Approach	Relationship	Engineering Rationale
AI Trustworthiness Governance	Trusted Operational Assurance	equivalent	Establishes governed oversight of AI model behavior in operational environments
Runtime AI Risk Mediation	Adaptive Governance	strongly-supports	Enables governance response to evolving AI behavior and operational risk
Explainable AI Governance	Explainable Operational Behavior	supports	Preserves reconstructable evidence of AI governance decisions

L-AI-002 — AI Output Validation

Technique	Approach	Relationship	Engineering Rationale
Trusted Output Validation	Operational Integrity Assurance	equivalent	Validates AI-generated outputs before downstream operational use
Runtime AI Governance Mediation	Runtime Governance Mediation	strongly-supports	Prevents invalid or unsafe AI outputs from influencing operational execution
Safe AI-Enabled Operations	Safe Operational Behavior	supports	Reduces risk of unsafe state transitions driven by AI-generated outputs

L-AI-003 — AI Confidence and Explainability Enforcement

Technique	Approach	Relationship	Engineering Rationale
Explainable AI Engineering	Explainable Operational Behavior	equivalent	Preserves confidence, rationale, and interpretability metadata for AI outputs
Trustworthiness Assessment	Trusted Operational Assurance	strongly-supports	Enables evaluation of AI output reliability and operational trustworthiness

Technique	Approach	Relationship	Engineering Rationale
Evidence-Aware AI Governance	Evidence-Centric Operations	supports	Maintains attributable evidence for AI-generated recommendations

L-AI-004 — Human Oversight for High-Impact AI Decisions

Technique	Approach	Relationship	Engineering Rationale
Supervisory AI Governance	Human-Governed Operations	equivalent	Requires human oversight for high-impact AI-enabled operational decisions
Elevated-Risk Execution Control	Safe Operational Behavior	strongly-supports	Prevents unsafe autonomous execution for high-impact actions
Accountability-Aware Authorization	Accountability Preservation	supports	Preserves attributable human approval and escalation records

L-AI-005 — AI Drift and Behavior Monitoring

Technique	Approach	Relationship	Engineering Rationale
AI Behavior Monitoring	Trusted Operational Assurance	equivalent	Continuously monitors AI behavior for drift and anomalous operational patterns
Adaptive Trust Assessment	Adaptive Governance	strongly-supports	Enables governance response to degraded AI trustworthiness
Resilient AI Operations	Resilient Operational Control	supports	Maintains operational resilience under changing AI behavior conditions

L-AI-006 — AI Assurance Evidence and Traceability

Technique	Approach	Relationship	Engineering Rationale
AI Assurance Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence for AI validation, oversight, and operational behavior
AI Lifecycle Traceability	Accountability Preservation	strongly-supports	Maintains reconstructable lineage for AI governance and assurance activities
Explainable AI Reconstruction	Explainable Operational Behavior	supports	Enables review and reconstruction of AI-enabled operational decisions

L-ORCH-001 — Security Workflow Orchestration

Technique	Approach	Relationship	Engineering Rationale
Governed Workflow Orchestration	Runtime Governance Mediation	equivalent	Coordinates operational security workflows through governed orchestration
Cross-Domain Operational Control	Trusted Operational Control	strongly-supports	Aligns security actions across multiple operational domains
Explainable Workflow Execution	Explainable Operational Behavior	supports	Enables reconstructable analysis of orchestrated security workflows

L-ORCH-002 — Cross-Domain Coordination

Technique	Approach	Relationship	Engineering Rationale
Coordinated Defense Engineering	Survivability Engineering	equivalent	Coordinates governance and security activity across enterprise domains
Cross-Domain Control Integration	Resilient Operational Control	strongly-supports	Reduces fragmentation and improves defensive coherence

Technique	Approach	Relationship	Engineering Rationale
Operational Dependency Awareness	Trusted Operational Awareness	supports	Preserves visibility into inter-domain governance relationships

L-ORCH-003 — Workflow Authorization Governance

Technique	Approach	Relationship	Engineering Rationale
Authorized Workflow Execution	Trusted Operational Control	equivalent	Ensures orchestrated workflows execute only under governed authorization
Runtime Authorization Mediation	Runtime Governance Mediation	strongly-supports	Prevents unauthorized orchestration across operational systems
Accountability-Aware Workflow Control	Accountability Preservation	supports	Maintains attributable workflow authorization history

L-ORCH-004 — Autonomous Workflow Constraint Enforcement

Technique	Approach	Relationship	Engineering Rationale
Constraint-Aware Autonomous Orchestration	Safe Operational Behavior	equivalent	Constrains autonomous workflow behavior within governed operational boundaries
Runtime Invariant Preservation	Runtime Governance Mediation	strongly-supports	Prevents orchestration from violating trusted operational constraints
Trusted Autonomous Coordination	Trusted Operational Control	supports	Maintains bounded authority for automated enterprise workflows

L-ANL-001 — Governance Metrics Collection

Technique	Approach	Relationship	Engineering Rationale
Governance Measurement Engineering	Trusted Operational Assurance	equivalent	Collects governance metrics for assurance and operational evaluation
Continuous Operational Assessment	Adaptive Governance	strongly-supports	Enables governance improvement using measured operational outcomes
Evidence-Aware Metrics Lineage	Evidence-Centric Operations	supports	Preserves attributable lineage for governance metrics and analytics

L-ANL-002 — Cross-Domain Telemetry Correlation

Technique	Approach	Relationship	Engineering Rationale
Cross-Domain Telemetry Engineering	Trusted Operational Awareness	equivalent	Correlates telemetry across governance, security, data, and infrastructure domains
Enterprise Situational Awareness	Resilient Operational Control	strongly-supports	Improves operational resilience through integrated telemetry visibility
Explainable Operational Correlation	Explainable Operational Behavior	supports	Enables reconstructable analysis of enterprise operational state

L-ANL-003 — Risk and Exposure Analytics

Technique	Approach	Relationship	Engineering Rationale
Operational Exposure Analytics	Adaptive Governance	equivalent	Produces analytics for enterprise risk, exposure, and governance posture
Dynamic Risk Evaluation	Resilient Operational Awareness	strongly-supports	Enables risk-aware adaptation of operational governance controls

Technique	Approach	Relationship	Engineering Rationale
Explainable Risk Assessment	Explainable Operational Behavior	supports	Preserves reconstructable rationale for exposure and risk analytics

L-ANL-004 — Operational Trend and Drift Detection

Technique	Approach	Relationship	Engineering Rationale
Operational Drift Detection	Trusted Operational Assurance	equivalent	Detects degradation, anomaly, or drift in operational and governance behavior
Adaptive Governance Triggering	Adaptive Governance	strongly-supports	Enables governance response to emerging operational trends
Trusted State Awareness	Trusted Operational Awareness	supports	Maintains visibility into deviation from trusted operational conditions

L-ASSUR-001 — Continuous Assurance Evaluation

Technique	Approach	Relationship	Engineering Rationale
Continuous Assurance Engineering	Trusted Operational Assurance	equivalent	Continuously evaluates operational assurance and governance posture
Runtime Trustworthiness Validation	Trusted Operational State	strongly-supports	Confirms that operational systems remain within trusted governance conditions
Evidence-Aware Assurance	Evidence-Centric Operations	supports	Preserves attributable assurance evaluation evidence

L-ASSUR-002 — Cross-Domain Assurance Correlation

Technique	Approach	Relationship	Engineering Rationale
Assurance Correlation Engineering	Trusted Operational Awareness	equivalent	Correlates assurance evidence across operational domains
Systemic Trust Evaluation	Resilient Operational Awareness	strongly-supports	Identifies assurance degradation across interdependent enterprise systems
Explainable Assurance Reconstruction	Explainable Operational Behavior	supports	Enables reconstructable review of assurance relationships and findings

L-ASSUR-003 — Assurance Drift Detection

Technique	Approach	Relationship	Engineering Rationale
Assurance Drift Detection	Adaptive Governance	equivalent	Detects degradation in governance and operational assurance posture
Runtime Trust Degradation Awareness	Trusted Operational Assurance	strongly-supports	Maintains visibility into declining operational trustworthiness
Resilient Governance Adaptation	Resilient Operational Control	supports	Enables adaptive remediation of assurance deficiencies

L-ASSUR-004 — Assurance Escalation and Remediation Coordination

Technique	Approach	Relationship	Engineering Rationale
Assurance Remediation Governance	Adaptive Governance	equivalent	Coordinates response to governance and assurance deficiencies

Technique	Approach	Relationship	Engineering Rationale
Supervisory Assurance Escalation	Human-Governed Operations	strongly-supports	Escalates high-impact assurance failures for accountable review
Controlled Assurance Recovery	Trusted Operational Control	supports	Preserves governed remediation of degraded assurance conditions

L-FEED-001 — Operational Feedback Collection

Technique	Approach	Relationship	Engineering Rationale
Operational Feedback Engineering	Adaptive Governance	equivalent	Collects operational findings for governance and control improvement
Continuous Learning Governance	Resilient Operational Control	strongly-supports	Enables system improvement based on observed operational behavior
Evidence-Aware Feedback Capture	Evidence-Centric Operations	supports	Preserves attributable feedback records and improvement signals

L-FEED-002 — Cross-Domain Learning Integration

Technique	Approach	Relationship	Engineering Rationale
Cross-Domain Learning Engineering	Adaptive Governance	equivalent	Integrates lessons across operational and governance domains
Coordinated Improvement Governance	Resilient Operational Control	strongly-supports	Improves defensive coherence across dependent enterprise functions
Explainable Learning Integration	Explainable Operational Behavior	supports	Preserves reconstructable rationale for cross-domain governance improvement

L-FEED-003 — Control Improvement Recommendation

Technique	Approach	Relationship	Engineering Rationale
Control Improvement Engineering	Adaptive Governance	equivalent	Recommends governance and control improvements from operational evidence
Assurance-Informed Optimization	Trusted Operational Assurance	strongly-supports	Improves control effectiveness based on measured assurance outcomes
Evidence-Linked Improvement	Evidence-Centric Operations	supports	Maintains traceability between operational findings and recommended changes

L-FEED-004 — Governance Adaptation and Optimization

Technique	Approach	Relationship	Engineering Rationale
Governance Adaptation Engineering	Adaptive Governance	equivalent	Continuously adapts governance workflows and operational controls
Resilient Governance Optimization	Resilient Operational Control	strongly-supports	Improves system resilience through adaptive control refinement
Explainable Governance Evolution	Explainable Operational Behavior	supports	Preserves rationale and evidence for governance adaptation

L-EVID-001 — Program Evidence and Governance Traceability

Technique	Approach	Relationship	Engineering Rationale
Program Evidence Preservation	Evidence-Centric Operations	equivalent	Produces attributable evidence for program governance and assurance activities

Technique	Approach	Relationship	Engineering Rationale
Enterprise Accountability Preservation	Accountability Preservation	strongly-supports	Maintains reconstructable security-program governance history
Explainable Program Reconstruction	Explainable Operational Behavior	supports	Enables review and reconstruction of enterprise governance evolution

17. Cross-Domain Systems Security Engineering Alignment Summary

The Practitioner’s Blueprint for Secure AI (PBSAI) Governance Ecosystem aligns comprehensively with the systems-security engineering principles described in NIST SP 800-160 Volume 2 through:

- deterministic runtime governance,
- trusted operational state preservation,
- survivability-oriented orchestration,
- explainable operational behavior,
- adaptive governance enforcement,
- bounded operational authority,
- provenance-aware operational control,
- evidence-centric accountability,
- and continuously validated operational assurance.

PBSAI operationalizes systems-security engineering directly within runtime enterprise operations rather than treating governance solely as:

- organizational policy,
- static architecture documentation,
- or post-execution compliance review.

The architecture embeds trustworthy systems-security engineering concepts directly into:

- operational execution pathways,
- orchestration workflows,
- authorization mediation,

- resilience coordination,
- telemetry analytics,
- AI-enabled decision workflows,
- and cross-domain governance enforcement.

17.1 Systems Security Engineering Alignment Summary Table

Systems Security Engineering Area	Primary PBSAI Domains	Operational Characteristics
Governance and Operational Control	A, C, E, L	Deterministic governance mediation, bounded authority, runtime authorization
Trusted Operational State	B, F, I, H	Trusted state preservation, integrity validation, secure restoration
Survivability and Resilience	G, H, I, K	Adaptive continuity, containment, recovery orchestration, dependency resilience
Trusted Operational Awareness	D, J, K, L	Cross-domain telemetry correlation, situational awareness, operational visibility
Explainable Operational Behavior	G, I, L	Explainable workflows, reconstructable decisions, operational transparency
Evidence-Centric Accountability	B, D, F, G, K, L	Provenance, traceability, attributable governance lineage
Adaptive Governance and Runtime Mediation	A, D, E, L	Dynamic governance adaptation, runtime orchestration constraints
Trusted Supply Chain and Lifecycle Security	K	Provenance validation, SBOM governance, lifecycle integrity assurance
AI Governance and Operational Assurance	L	AI validation, HITL governance, assurance drift detection

17.2 PBSAI Trustworthiness Engineering Characteristics

Engineering Characteristic	PBSAI Alignment Strength	Primary Supporting Capabilities
Trusted Operational State	Very Strong	Configuration governance, integrity validation, recovery assurance
Runtime Governance Mediation	Very Strong	Deterministic authorization, orchestration constraints, policy enforcement
Survivability Engineering	Very Strong	Containment, segmentation, recovery orchestration, failover governance
Adaptive Governance	Very Strong	Drift detection, feedback integration, threat-adaptive controls
Evidence-Centric Operations	Very Strong	Telemetry evidence, provenance, operational lineage reconstruction
Explainable Operational Behavior	Strong	Structured operational evidence, explainability metadata, reconstruction workflows
Trusted Operational Assurance	Very Strong	Continuous monitoring, assurance correlation, validation workflows
Safe Operational Behavior	Very Strong	Bounded remediation, HITL constraints, rollback governance
Trusted Operational Awareness	Very Strong	Cross-domain telemetry fusion, situational awareness, threat intelligence
Accountability Preservation	Very Strong	Identity-linked governance, evidence continuity, attributable operational history

17.3 Runtime Governance and Survivability Differentiators

PBSAI differs from traditional systems-security engineering approaches by operationalizing:

- runtime governance mediation,
- continuously adaptive orchestration governance,
- explainable operational authorization,

- deterministic execution constraints,
- AI-aware operational assurance,
- and evidence-centric operational survivability.

Key differentiators include:

PBSAI Capability	Operational Significance
Deterministic Runtime Governance	Prevents unauthorized operational execution before state transition
Context Envelope Governance	Preserves governance context throughout operational workflows
Structured Output Contracts	Enables deterministic validation of AI-generated operational outputs
Bounded Autonomous Execution	Restricts unsafe orchestration and automated operational behavior
Provenance-Aware Operations	Maintains attributable operational lineage and trust continuity
Cross-Domain Telemetry Correlation	Provides enterprise-scale situational awareness and operational reconstruction
Adaptive Governance Enforcement	Dynamically adjusts governance according to operational conditions
Recovery and Rollback Governance	Enables trusted restoration following unsafe or failed operational transitions
Continuous Assurance Validation	Maintains ongoing evaluation of operational trustworthiness
HITL Escalation Governance	Preserves human authority for high-impact operational decisions

17.4 PBSAI as a Systems-Security Operationalization Architecture

PBSAI demonstrates how systems-security engineering principles can evolve from:

- conceptual engineering guidance,
into:

- continuously operating runtime governance systems.

The architecture operationalizes:

- trustworthy operational state management,
- deterministic execution governance,
- resilient orchestration,
- explainable operational workflows,
- survivability-aware runtime control,
- and continuously adaptive operational assurance.

PBSAI therefore functions simultaneously as:

- a runtime governance architecture,
- a resilient orchestration governance plane,
- a trustworthy AI operationalization framework,
- and a systems-security engineering realization architecture for AI-enabled enterprise systems.

The framework demonstrates how enterprise systems can preserve:

- operational trustworthiness,
- survivability,
- accountability,
- explainability,
- resilience,
- and bounded autonomy,
even under:
- adversarial conditions,
- evolving operational risk,
- dynamic orchestration workflows,
- and AI-enabled autonomous execution environments.

18. Conclusion

The Practitioner's Blueprint for Secure AI (PBSAI) Governance Ecosystem provides a comprehensive systems-security operationalization architecture for AI-enabled enterprise environments.

PBSAI aligns strongly with the systems-security engineering principles described in NIST SP 800-160 Volume 2 through:

- deterministic runtime governance,
- trusted operational state preservation,
- explainable operational mediation,
- bounded autonomous execution,
- survivability-oriented orchestration,
- evidence-centric accountability,
- adaptive governance enforcement,
- and continuously validated operational assurance.

Unlike traditional governance approaches focused primarily on:

- static compliance,
- organizational process management,
- or post-execution auditing,

PBSAI embeds governance directly into runtime operational execution.

The architecture operationalizes:

- policy-aware orchestration,
- trusted operational mediation,
- context-preserving governance,
- explainable execution control,
- provenance-aware operational trust,
- resilient recovery governance,
- and adaptive survivability-oriented security engineering.

PBSAI therefore serves simultaneously as:

- a secure AI governance architecture,

- a systems-security operationalization framework,
- a resilient orchestration governance plane,
- and a runtime trustworthy systems engineering realization model.

The architecture demonstrates how AI-enabled enterprise systems can evolve from:

- passive governance compliance models, into:
- continuously governed operational ecosystems capable of preserving trustworthiness, survivability, explainability, accountability, and resilience throughout runtime enterprise operations.

Appendix A — Relationship Semantics

Relationship	Definition
equivalent	PBSAI materially realizes the systems-security engineering technique and approach
strongly-supports	PBSAI substantially enables the engineering objective
supports	PBSAI partially contributes to the engineering objective
operationalizes	PBSAI converts engineering guidance into runtime governance enforcement
constrains	PBSAI enforces runtime engineering boundaries
extends	PBSAI introduces governance capabilities beyond the engineering guidance

Appendix B — Systems Security Engineering Characteristic Definitions

Characteristic	Definition
Trusted Operational State	Ability to maintain verifiable trustworthy runtime conditions
Runtime Governance Mediation	Deterministic governance enforcement during operational execution

Characteristic	Definition
Survivability Engineering	Ability to preserve operational continuity under adverse conditions
Adaptive Governance	Ability to evolve governance behavior according to operational conditions
Evidence-Centric Operations	Production of attributable operational evidence and traceability
Explainable Operational Behavior	Ability to reconstruct and interpret operational decisions and workflows
Trusted Operational Assurance	Continuous validation of operational trustworthiness
Safe Operational Behavior	Prevention of unsafe or uncontrolled operational execution
Trusted Operational Awareness	Continuous situational visibility into operational conditions
Accountability Preservation	Ability to attribute actions, decisions, and governance activities

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